

Upscaling Hydraulic Properties and Soil Water Flow Processes in Heterogeneous Soils: A Review

H. Vereecken,* R. Kasteel, J. Vanderborght, and T. Harter

ABSTRACT

This review covers, in a comprehensive manner, the approaches available in the literature to upscale soil water processes and hydraulic parameters in the vadose zone. We distinguish two categories of upscaling methods: forward approaches requiring information about the spatial distribution of hydraulic parameters at a small scale, and inverse modeling approaches requiring information about the spatial and temporal variation of state variables at various scales, including so-called “soft data”. Geostatistical and scaling approaches are crucial to upscale soil water processes and to derive large-scale effective fluxes and parameters from small-scale information. Upscaling approaches include stochastic perturbation methods, the scaleway approach, the stream-tube approach, the aggregation concept, inverse modeling approaches, and data fusion. With all upscaling methods, the estimated effective parameters depend not only on the properties of the heterogeneous flow field but also on boundary conditions. The use of the Richards equation at the field and watershed scale is based more on pragmatism than on a sound physical basis. There are practically no data sets presently available that provide sufficient information to extensively validate existing upscaling approaches. Use of numerical case studies has therefore been most common. More recently and still under development, hydrogeophysical methods combined with ground-based remote sensing techniques promise significant contributions toward providing high-quality data sets. Finally, most of the upscaling literature in vadose zone research has dealt with bare soils or deep vadose zones. There is a need to develop upscaling methods for real world soils, considering root water uptake mechanisms and other soil–plant–atmosphere interactions.

Upscaling of soil water processes is one of the most important issues in vadose zone research and soil science. Sustainable management of the vadose zone environment as well as the protection of groundwater resources requires an in-depth understanding of the governing soil water flow processes as well as a characterization of its hydraulic properties. There exists, however, a large discrepancy between the relatively small, “local” scale at which soil water fluxes, vadose zone state variables, and vadose zone properties are usually measured, e.g., in soil cores, soil tensiometers, or with soil solution samplers, and the much larger field scale, watershed scale, or regional scale at which hydrologic management decisions are made. For predicting and understanding flow, solute, and energy fluxes at the field, watershed, and global climate modeling scale, vadose zone models

must be constructed at scales that are orders of magnitude larger than the scale at which vadose zone properties are determined.

Significant intrinsic (geologic and pedologic) and extrinsic (boundary-driven) spatial variability of vadose zone properties and processes are known to exist at the sub-Darcian pore scale (variations in pore properties across distances of few micrometers), and at the so-called “Darcy” or “local” scale of most laboratory and field instrument methods. Within the vertical soil profile, this scale typically encompasses a few millimeters to centimeters and laterally a few centimeters up to 1 m. Larger scale variations occur due to sedimentary layering vertically (decimeter to meter scale) and between soil mapping units laterally (meter to hectometer scale). Larger scale lateral variability also occurs due to spatial variations in the boundary fluxes across the soil–plant–atmosphere interface at the top of the vadose zone. Such variability is due to variability in precipitation, wind, and moisture flux across a wide range of scales, due to variability in agricultural management practices within and between individual fields, due to land surface geomorphic features, and due to spatial variability in the ecology of the land surface.

In contrast, the grid resolution of watershed, regional, and global climate models is on the order of hectometers to many kilometers laterally. What is the effective mathematical description of vadose zone processes at that scale and how are the parameters of such large-scale models being determined from local-scale information? The pursuit of this question is the focus of so-called upscaling procedures (in the broadest sense). Upscaling techniques provide either effective equations, or effective parameters, or both to describe the vadose zone system’s behavior (water, solute, and energy fluxes) at a scale that is relevant for management, assessment, and decision making, while capturing information collected at a much smaller scale and while accounting for the influence of heterogeneities at a scale not resolved by the model (Grayson and Blöschl, 2000; Harter and Hopmans 2004).

In this review, we have defined *upscaling* as the process that replaces a heterogeneous domain with a homogeneous one in such a manner that both domains produce the same response under some upscaled boundary conditions (Rubin, 2003). The difficulty in upscaling stems from the inherent spatial variability of soil properties and their often nonlinear dependence on state variables. The parameters describing the flow processes in the upscaled homogeneous medium are defined as *effective parameters*.

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Abbreviations: ERT, electrical resistivity tomography; GPR, ground-penetrating radar; PTF, pedotransfer function; TDR, time domain reflectometry.

Often a distinction is made between effective and equivalent parameters. Neuman and Di Federico (1998) defined effective parameters as those parameters that are used in ensemble-averaged equations (e.g., effective hydraulic conductivity relating the ensemble average flux to the ensemble mean gradient) and equivalent parameters that are present in spatially averaged equations. Practically speaking, *effective parameters* are an intrinsic property of the homogenized domain (not a function of the particular boundary conditions imposed on the domain), while *equivalent parameters* are those valid only for a specific set of boundary conditions. In this review we have used the term *effective* in a more general manner that includes the more specific meaning of *equivalent* as we refer to the upscaled properties and equations.

In upscaling soil water processes in the vadose zone, four different scales are typically distinguished in the literature (Harter and Hopmans, 2004) (Fig. 1). At the pore scale level, water flow is described by either the Stokes or Navier Stokes equations. Description of water flow at this scale requires a complete characterization of the pore geometry of the flow domain, phase boundary conditions, and fluid properties. The next scale is mostly referred to as the local, macroscopic, or Darcy scale. At this scale, Darcy's equation (momentum conservation) and Richards' equation (mass conservation) are typically used to describe soil water processes under variably saturated conditions. The next larger scale is the field scale, which is both a management unit (in agriculture) with its own decision space and also a typical experi-

mental unit to conduct vadose zone research. At that scale, Richards' equation is extensively used to describe and predict variably saturated water flow. Nielsen et al. (1973) were among the first to report on spatial variability of soil hydraulic properties and soil moisture content within an apparently homogeneous field. They found that even seemingly uniform land areas manifest large variations in hydraulic conductivity, while variations in water content, bulk density, and water content are clearly smaller. The study of Nielsen et al. (1973) initiated a large number of field-scale studies exploring the spatial variability of soil water processes and their related properties and state variables (see below). The fourth scale is the regional scale, corresponding to the catchment (watershed) scale in hydrology. At that scale, vadose zone processes are often represented by simple zero-order, black-box, lumped parameter models as part of a complex hydrological system that also accounts for relief, vegetation, weather or climate forces, surface runoff, flood routing, and evapotranspiration, among others.

There is an extensive body of literature on the upscaling of flow processes in porous media with applications to various disciplines such as petroleum and reservoir engineering, hydrogeology, hydrology, and soil science (e.g., Gelhar, 1993; Dagan, 1989; Renard and de Marsily, 1996; Cushman et al., 2002; Farmer, 2002; Sposito, 1998; Zhang, 2002; Pachepsky et al., 2003; Harter and Hopmans, 2004; Das and Hassanizadeh, 2005). A comprehensive description of all methods is beyond the scope of this review. Instead, we have focused on meth-

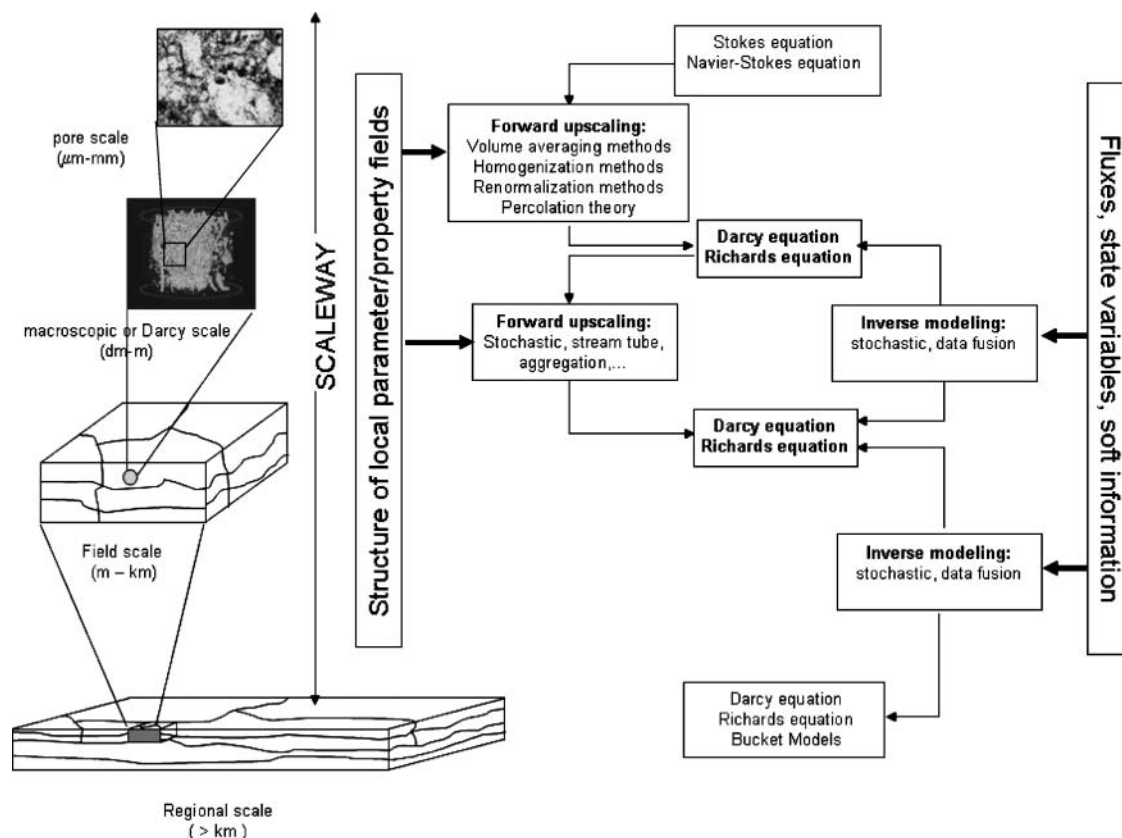


Fig. 1. Forward and inverse upscaling methods to upscale soil water flow processes from the local to the field scale.

ods that deal specifically with upscaling from the local (Darcy) scale of typical laboratory and field measurements (a few centimeters to decimeters) to an effective representation of unsaturated flow processes at the field and catchment scale (hectometers to kilometers). We discuss the strengths and the limitations of each of the approaches with specific attention to their data requirements and their potential for field applications. It is not our intention to analyze in full detail the advances made within each of the different conceptual frameworks but rather to give an overall view on what is presently available to estimate field-scale behavior of soil water flow processes as a way to derive an outline for future research needs.

We distinguish two major categories of upscaling procedures among those presently available to derive effective hydraulic properties and soil water fluxes. The first group contains methods that derive the parameters of the larger scale process model directly from explicit information about the spatial structure and variability of soil hydraulic properties measured at the small (local) scale. Some of these methods, such as stochastic perturbation and volume averaging, explicitly derive upscaled flow equations with effective hydraulic parameters. Using a forward upscaling model, larger scale properties and fluxes are determined. Subsequently, these approaches will be referred to as *forward upscaling* (Fig. 1).

The second group, called *inverse upscaling*, exploits not only information about the controlling system variables (i.e., model parameters, as in the forward approach), but also uses information contained in the measurement of state variables of soil water processes (e.g., soil water tension, water content) and knowledge about boundary fluxes (e.g., totalized recharge, tile drainage). Inverse models estimate model system parameters at the larger scale directly from an assortment of available information or infer information about the small scale (local) spatial structure of the parameter field or both. This involves a model inversion and therefore a priori assumptions about the large-scale process model and about the structure of the small (local) scale parameter distribution.

Some recent approaches expand on the classic inverse approach by further constraining the upscaling problem with secondary, indirect information about primary system and state variables. Such indirect information about, for example, hydraulic properties at the local scale can be obtained from geophysical measurements, remote sensing data, soil maps, and hydrogeological data (so-called "soft data"). This combined inverse modeling approach of using direct and indirect (secondary) measurements of state and system variables will be referred to as *data fusion* (Yeh and Simunek, 2002).

The review is organized in the following manner: We first give an overview of methods used to characterize spatial variability of local-scale properties. Next, we address the forward upscaling or parameter-driven approach (from the local to the field scale). This includes the three-dimensional stochastic perturbation approach, the concept of the scaleway approach including numerical upscaling, the stream-tube approach, the concept of aggregation, and a very brief overview of other rele-

vant methods such as volume averaging, homogenization, and renormalization. Then we deal with inverse upscaling methods, which make use of large-scale-averaged state values including inverse modeling of large-scale-averaged state values, analysis-of-moment equations that relate spatial moments of parameters and variables, and a geostatistically based estimation of the parameter field from the observed variable fields. This includes a review of data fusion to quantify spatial variability of soil moisture status by using geophysical and remote sensing methods.

CHARACTERIZING SPATIAL VARIABILITY AT THE LOCAL SCALE

The basis for any forward upscaling procedure is a mathematical-conceptual description of the spatial distribution of hydraulic or other vadose zone properties at some small scale at which measurements are taken. Characterization of the spatial variability of properties related to soil water (soil moisture retention characteristic and unsaturated hydraulic conductivity) is typically done at the local scale. For this purpose, small soil cores (centimeter-decimeter scale) are taken in the field and analyzed in the laboratory for their hydraulic properties using standard soil physical techniques. Spatial distributions of hydraulic properties and state variables are determined by taking multiple, closely spaced measurements across a site (Wierenga et al., 1991; Onsoy et al., 2005). A two-step approach is commonly used to describe spatial variability in the vadose zone for flow modeling purposes: Geostatistical methods are used to characterize the structure of spatial variability by use of formal statistical quantities that also define the spatio-temporal relationship between multiple physical variables, e.g., saturated hydraulic conductivity, air-entry pressure, and soil tortuosity factor. Scaling methods are applied before the geostatistical analysis to avoid dealing with multiple, spatially variable but correlated, physical quantities. Scaling methods reduce a multidimensional parameter space into a single-parameter space by taking advantage of the strongly linearizable relationship between soil hydraulic properties and the variable pore-space length scale.

Geostatistical Approaches

The application of geostatistical approaches requires that the hydraulic parameters be described as a function of location. Because it is generally impossible to define or predict parameters at every location in a deterministic sense, their spatial distribution is determined within a geostatistical framework. The geostatistical information is then used as part of an upscaling procedure (e.g., stochastic method) to predict effective hydraulic parameters, upscaled state variables, and fluxes. Geostatistical characterization of a spatially variable quantity is based on the theory of regionalized variables (Matheron, 1971; Journel and Huijbregts, 1978). We will give a brief outline following the description of Russo and Jury (1987a, 1987b). Within a geostatistical framework, a parameter v , e.g., saturated hydraulic conductivity, is considered a

random realization of a (three-dimensional) random space function (RSF) $V(\mathbf{x})$, which is defined by a structural or so-called “geostatistical” model (Russo and Jury, 1987a, 1987b) that includes a deterministic and a random, but spatially correlated, component:

$$V(\mathbf{x}) = m(\mathbf{x}) + \varepsilon(\mathbf{x}) \quad [1]$$

where $\mathbf{x}$ is the spatial coordinate vector, $m(\mathbf{x})$ is the deterministic drift function or prior mean, and $\varepsilon(\mathbf{x})$ is the RSF component. A complete characterization of the spatially correlated, random component $\varepsilon(\mathbf{x})$ would involve the definition of an infinite number of probability density functions (pdfs) $f^n(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$ which represents the n -point pdf of $\varepsilon(\mathbf{x})$ at all locations $\mathbf{x}_1$ to $\mathbf{x}_n$. Since this is not possible, generally two simplifying assumptions are made. The first is the assumption of stationarity, which means that the pdfs are translation invariant: $f^n(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) = f^n(\mathbf{x}_1 + \mathbf{h}, \mathbf{x}_2 + \mathbf{h}, \dots, \mathbf{x}_n + \mathbf{h})$. The second is that the first two statistical moments are invariant under spatial translation. The entire RSF is therefore defined by the two-point joint pdf and a complete characterization of the random variable is achieved by describing its first and second spatial moments:

$$E[\varepsilon(\mathbf{x})] = 0 \quad [2]$$

and

$$C(\mathbf{h}) = E[\varepsilon(\mathbf{x})\varepsilon(\mathbf{x} + \mathbf{h})] \quad [3]$$

where $C(\mathbf{h})$ is the spatial covariance of ε for a lag distance $\mathbf{h}$.

Typically, the spatial structure of the stochastic functions $V(\mathbf{x})$ is defined either by the so-called semivariogram or by the correlogram. The semivariogram is given by

$$2\gamma(\mathbf{h}) = E\{[V(\mathbf{x} + \mathbf{h}) - V(\mathbf{x})]^2\} \quad [4]$$

and the correlation function or correlogram by

$$\rho(\mathbf{h}) = C(\mathbf{h})/C(0) = C(\mathbf{h})/E[(V(\mathbf{h}) - m)^2] \quad [5]$$

where m is the mean value. The application of stochastic upscaling methods (see below) requires information about the spatial structure of the complete moisture retention characteristic and the unsaturated hydraulic conductivity function. These functions are described by a parameter vector. Its dimension depends on the type of model selected. To apply stochastic perturbation theories, the spatial structure of the individual parameters, but also cross-covariances between the parameters and state variables, needs to be derived. Again, second-order stationarity is typically assumed. Then a cross-covariance function between two stochastic functions $V(\mathbf{x})$ and $U(\mathbf{x})$ is given by

$$C_{v,u}(\mathbf{h}) = E\{[V(\mathbf{x} + \mathbf{h}) - m_v][U(\mathbf{x}) - m_u]\} \quad [6]$$

where the subscripts $v(\mathbf{x})$ and $u(\mathbf{x})$ refer to stochastic functions of two parameters, e.g., air-entry pressure and saturated hydraulic conductivity, and m_i refers to the respective mean values of the stochastic functions.

Geostatistical concepts have been used in many studies to analyze field data of soil hydraulic properties. Most of these studies have focused on deriving the sta-

tistical properties of single-parameter distributions (Nielsen et al., 1973; Wierenga et al., 1991; Khaleel and Relyea, 2001), correlations between hydraulic parameters (e.g., Beyers and Stephens, 1983) and semivariograms or covariance functions (e.g., Russo and Bresler, 1981; Jury et al., 1987; Jensen and Refsgaard, 1991; White and Sully, 1992; Russo and Bouton, 1992; Russo et al., 1997; Shouse et al., 1995; Mallants et al., 1996; Vereecken et al., 1997; de Rooij et al., 2004). Some of the studies paid specific attention to the presence of statistical anisotropy in hydraulic parameters (e.g., Russo and Bouton, 1992; Russo et al., 1997), statistical methods to detect and analyze spatial structures (Russo and Jury, 1987a, 1987b; Jury et al., 1987; Ünlü et al., 1990), and the presence of nonstationary behavior of soil hydraulic parameters (Jury et al., 1987; Russo and Bouton, 1992; Russo et al., 1997). It should be noted that information on the spatial correlation of hydraulic properties in the vertical direction is much less common than information on lateral correlation, although the spatial correlation structure of hydraulic parameters in the direction of mean flow has a critical control on flow and transport processes. Moreover, field sites for which spatial correlation lengths of hydraulic properties could be derived are typically located in more arid regions and often in poorly developed soils with a coarser texture (e.g., Entisols) (Ünlü et al., 1990; Russo and Bouton, 1992; White and Sully, 1992; Rockhold et al., 1996). A similar number of studies show variograms of hydraulic parameters with a large nugget, indicating that no spatial structure could be identified (Mohanty et al., 1994; Mallants et al., 1996; Kasteel, 1997; Vereecken et al., 1997; Hammel et al., 1999; Deurer et al., 2000).

Only a few studies analyzed the spatial cross-covariances and semivariances of hydraulic parameters. Ünlü et al. (1990) used the Gardner equation:

$$K(\psi, \mathbf{x}) = K_s(\mathbf{x})\exp[\alpha(\mathbf{x})\psi(\mathbf{x}, t)] \quad [7]$$

where $K(\psi, \mathbf{x})$ ($L T^{-1}$) is the hydraulic conductivity, $\mathbf{x}$ the spatial coordinate, t time, $\psi(\mathbf{x}, t)$ the pressure head, and $\alpha(\mathbf{x})$ a pore size distribution parameter to describe measured moisture retention characteristics and unsaturated hydraulic conductivity obtained from neutron probe measurements from a loam soil. They observed a positive exponential cross-correlation between $\ln(K_s)$ and α . Vauclin et al. (1994) measured the unsaturated hydraulic conductivities in a loamy soil using the Guelph permeameter and derived the spatial statistics of K_s and α in the Gardner equation as well as their cross-covariance. They found a negative spatial correlation between $\ln(K_s)$ and $\ln(\alpha)$ at distances up to 24 m.

Mallants et al. (1996) analyzed the spatial variability of the van Genuchten parameters θ_s , θ_r , α_{vg} , and n (van Genuchten, 1980) and the saturated hydraulic conductivity using 180 soil cores along a 31-m-long transect in a multilayered soil profile. In addition to the univariate geostatistical analysis, they also derived cross-covariances between the various parameters. They found that the correlation scales of cross-covariances were of a similar magnitude as those pertaining to the direct variograms of cross-correlated parameters, and were typically on the order of 5 to 10 m.

Parameters that describe unsaturated hydraulic properties are not typically measured directly but derived from in situ experimental data using inverse modeling (e.g., data from multistep outflow experiments) or using approximate analytical solutions of the flow equation (e.g., data from infiltration experiments). High sensitivity to measurement errors and potential hydraulic model errors lead to uncertainty about the real hydraulic parameter values and to artificial spatial correlation among estimated hydraulic parameters. To what extent the observed variability and spatial correlation structure of estimated hydraulic parameters represents true structural variability versus measurement uncertainty has received little attention. A study by Mertens et al. (2002) demonstrated that parameter uncertainty may be responsible for a significant part of the observed variability of the hydraulic parameters.

In summary, few field data sets presently exist that provide information on the full spatial structure of the moisture retention characteristic and the unsaturated hydraulic conductivity function including the cross-covariances between hydraulic parameters. Data sets that additionally provide the spatial and temporal statistical structure of state variables to validate upscaling theories of soil water processes are even rarer. This is due to the high number of samples needed to characterize the spatial structure, the investment in time and money needed to determine the hydraulic properties despite novel methods like the multistep outflow method, and the costs involved in setting up this type of field experiments.

The Scaling Approach

The scaling approach was introduced in soil science by the pioneering work of Miller and Miller (1956). They derived scaling relationships for matrix potential or pressure head, ψ , and hydraulic conductivity, K , among others, using the microscopic laws for capillary pressure forces and viscous flow in porous media based on a similarity assumption of the pore space structure. Scale factors in general are defined as conversion factors that relate the characteristics of one system to the corresponding characteristics of another system (Tillotson and Nielsen, 1984). These researchers distinguished two approaches in deriving scaling factors of soil water processes: (i) the dimensional analysis, called *similitude analysis*, which is based on the physical characteristics of a soil, and (ii) the functional normalization approach, which is based on regression analysis. The dimensional analysis approach directly refers to the theory developed by Miller and Miller (1956) and includes fractal-based approaches (Tyler and Wheatcraft, 1990, 1992; Rieu and Sposito, 1991a, 1991b, 1991c; Perrier et al., 1996; Pachepsky et al., 2000). Several examples of applying dimensional techniques to soil properties are given in Tillotson and Nielsen (1984). Sposito (1990, 1998), following up on the work of Miller and Miller, studied the invariance of the Richards equation under scaling transformations. Recently, Williams and Ahuja (2003) and Kozak and Ahuja (2005a) proposed a scaling approach to determine the moisture retention charac-

teristics and hydraulic conductivity curve for a broad range of textural classes based on the work of Gregson et al. (1987) and Ahuja and Williams (1991). Using this approach, Kozak et al. (2005) attempted to scale and estimate evaporation and transpiration of water across soil textures. We briefly review dimensional analysis and the functional renormalization approach. Both approaches have been commonly used to reduce the number of spatially dependent parameters, thereby facilitating geostatistical characterization of heterogeneity in heterogeneous vadose zones.

Scaling of Hydraulic Properties

In the scaling theory developed by Miller and Miller (1956), two soils or porous media are similar when scale factors can be found that transform the behavior of one of the porous media to that of the other (Nielsen et al., 1998). The theory is founded on the concept that the geometry of pore spaces at any location, $\mathbf{x}$, are identical to one another after transforming it to a reference state with a microscopic characteristic length, χ^* . The variability of the pore space can be represented by characterizing the variability in microscopic characteristic length, χ , and relating it to the hydraulic properties:

$$K(\mathbf{x}, \theta) = K^*(\theta) \lambda_I(\mathbf{x})^2 \quad [8]$$

$$\psi(\mathbf{x}, \theta) = \lambda_I(\mathbf{x})^{-1} \psi^*(\theta) \quad [9]$$

where $\lambda_I = \chi(\mathbf{x})/\chi^*$ are the scale factors at location $\mathbf{x}$ having a mean value of unity and $K^*(\theta)$ and $\psi^*(\theta)$ are the reference hydraulic functions describing the reference state of the soil at volumetric water content θ . Geometric similarity also implies that the scaling factors are identical for both the pressure head and the hydraulic conductivity. Miller similarity is defined at the microscopic scale of the pore space and requires that porous media at different locations differ only in the scale of internal microscopic geometry and therefore have equal porosity. Laboratory studies have shown that the stringent criteria for Miller similarity are met for repacked quartz sand columns (Klute and Wilkinson, 1958; Schroth et al., 1996).

Another approach to scale hydraulic properties was developed by Leverett (1941). He found that experimental soil water retention characteristics from unconsolidated sands could be scaled to one single curve, called the J function, by using the following normalization:

$$J(S_e) = \frac{\psi}{\sigma} \sqrt{\frac{k}{\theta_s}} \quad [10]$$

where S_e is the effective saturation defined as $(\theta - \theta_r)/(\theta_s - \theta_r)$, with θ_r and θ_s the residual and saturated water content, respectively, k the intrinsic permeability, and σ the surface tension. Through this type of scaling, the moisture retention characteristic of one medium can be related to the moisture retention characteristic of the reference medium containing the same liquid by using

$$\psi(S_e) = \sqrt{\frac{K_{\text{ref}} \theta_s^{\text{ref}}}{K \theta_s}} \psi^{\text{ref}}(S_e) \quad [11]$$

where the subscript and superscript *ref* refer to the reference medium. In the case of uniform porosity and using a Leverett scaling relation equivalent to Eq. [9], Eq. [11] yields

$$K = \lambda_{L,ref}^2 K_{ref} \quad [12]$$

with $\lambda_{L,ref}$ the Leverett scaling factor equal to $\sqrt{(K/K_{ref})}$. Notice that Eq. [12] differs from Eq. [8] only in the definition of the scaling factors. In practice, scaling of the soil water retention curve and application of those scaling factors to the unsaturated hydraulic conductivity is referred to as *Miller–Miller type scaling*. Scaling of the unsaturated hydraulic conductivity curve (Eq. [12]) and application of those scaling factors to the soil water retention curve is referred to as *Leverett type scaling* (Oliveira et al., 2006).

Warrick et al. (1977) analyzed the experimental data of Nielsen et al. (1973) using scaling theory and found that the restriction of constant porosity, implicitly contained in the concept of Miller and Miller (1956), could not be maintained due to the large variability in total porosity. This restriction was relaxed by introducing the degree of effective water saturation, S_e . This implies that the water content can be scaled relative to its value at saturation. Figure 2 shows an example of using saturation to scale moisture retention characteristics observed in the field. A total of 154 soil cores were taken to a depth of 1.2 m from four adjacent soil profiles in a Typic Eutrochrept, which has three distinct layers (Kasteel, 1997). Scaling reduced the variation by 81%. The scaling relationships for the hydraulic properties of these so-called *Warrick-similar soils* (see e.g., Warrick and Nielsen, 1980) are formally similar to the corresponding expressions for Miller similitude after replacing θ with S_e .

Following the similar-media concept, and thus implicitly assuming equal porosity, Kosugi and Hopmans (1998) proposed a physically based scaling method based on the assumption that the soil pore radius is log-normally distributed in the study area. Hence, scaling factors are also log-normally distributed. This is consistent with evidence from field data, which has repeatedly shown, for example, log-normally distributed saturated hydraulic conductivity. Kosugi and Hopmans (1998) defined the mean pore-size radius, for which the effective saturation equals 0.5, as the microscopic characteristic length scale.

Scaling factors and parameters of the reference curve were computed directly from the parameters that fitted the individual measured soil moisture curves using the model proposed by Kosugi (1996), as opposed to conventional scaling in which scaling factors are estimated by minimizing the residual sum of square differences between the data and the scaled mean curve. Tuli et al. (2001) extended the approach by Kosugi and Hopmans (1998) to the simultaneous scaling of soil water retention and the hydraulic conductivity curve.

The fractal concept of Mandelbrot (1983) also belongs to the class of scaling methods and has been used to study vadose zone processes. The three characteristics of fractal patterns in the vadose zone are (Perrier et al., 1996): (i) they possess a similar structure across a range of length scales; (ii) their structure is scale independent; and (iii) the structure or pattern cannot be captured entirely by classical geometrical concepts. Tyler and Wheatcraft (1990) showed that the water retention characteristic of a fractal porous medium can be characterized as a so-called Sierpinski carpet. They also demonstrated that the exponent in the Brooks–Corey soil water retention equation (Brooks and Corey, 1964) effectively represents a fractal number that defines the (fractal) distribution of soil pore sizes. Rieu and Sposito (1991a, 1991b, 1991c) further developed a fully self-consistent fractal model of aggregate and pore-space properties for structured soils that includes both a fractal and a nonfractal structure component. A generalized model of both these earlier works was presented by Perrier et al. (1996). Fractal vadose zone structure is consistent with field evidence of power-law variograms of, for example, depth-averaged spatiotemporally varying soil moisture across a 65-ha field site (Green and Erskine, 2004).

The second main group of methods to determine scaling factors is the functional normalization approach, which uses least square regression analysis to derive scale factors from a set of experimental observations. This method is referred to as *functional normalization*. The objective of functional normalization is to coalesce all measured values of a relationship, e.g., between ψ and θ or K and θ , into one reference curve using scaling relations as defined in Eq. [8] and [9]. One single scaling factor is used for each soil sample and for each hydraulic

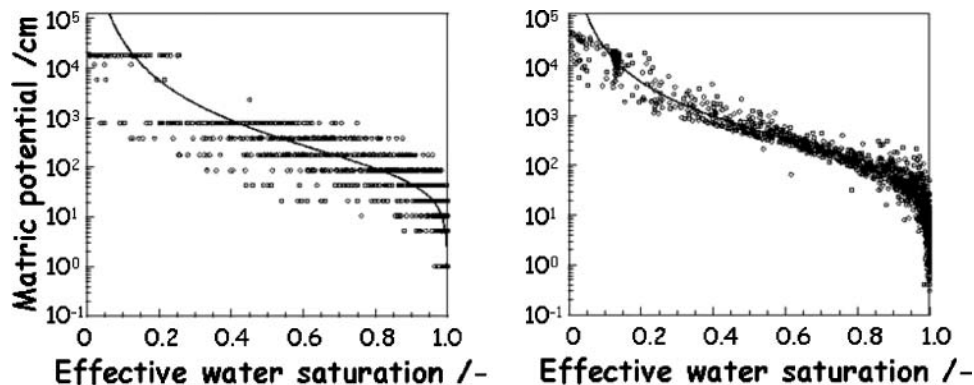


Fig. 2. Scaling of measured soil moisture characteristic curves at the field site using effective saturation (data from Kasteel, 1997).

relationship. This yields a frequency distribution of scaling factors. The scaling factors are interpreted in the same way as those derived by dimensional analysis, although they have been obtained without any physical justification. Various methods have been proposed in the literature to determine the scaling factors for matric potential and hydraulic conductivity (Peck et al., 1977; Warrick et al., 1977; Simmons et al., 1979; Russo and Bresler, 1980; Vogel et al., 1991). Hopmans (1987) compared the scaling factors obtained with various methods. These studies showed that the scaling factors for pressure head and hydraulic conductivity have a log-normal rather than a normal distribution. Since the scaling factors obtained using the functional normalization approach are not based on a physical model or certain assumptions about the self-similar structure of the porous medium, there is no constraint on the relation between the scaling factors for the $K(\theta)$ and $\psi(\theta)$ functions. Warrick et al. (1977) found that the values of α needed for scaling ψ differed from those for scaling K . These findings were corroborated in many other field studies (e.g., Ahuja et al., 1984, 1989; Hills et al., 1991).

The previously mentioned methods scale the hydraulic functions independently, with separate scaling factors for the water retention curve and hydraulic conductivity function. Clausnitzer et al. (1992) introduced a method to scale both hydraulic functions simultaneously, yielding a single set of scaling factors that then is consistent with the Miller similitude scaling relationships. They have shown that simultaneous scaling was not always as successful as independent scaling, whereas Hendrayanto et al. (2000) found that simultaneous scaling was the best among five methods tested to scale the hydraulic properties of forest soils.

Vogel et al. (1991) reformulated and extended the concept of similar media to analyze unsaturated or saturated flow in a system of parallel, nonhomogeneous, one-, two-, and three-dimensional soil profiles. Their linear variability concept is based on the functional similarity, which describes the similarity between soil hydraulic properties instead of similarity between internal geometries (Simmons et al., 1979). The total variability in soil hydraulic properties is thought to be composed of a linear and a nonlinear component. The nonlinear component depends on texture and is characterized by the shape of the hydraulic functions. Their analysis also accounts for the time variability of hydraulic properties. Deurer et al. (2000) used soil horizons as “functional similar units”. They found that linear and nonlinear scaling factors showed similar spatial structure in the horizontal but not in the vertical direction, with no cross-correlation between the scaling factors.

Oliveira et al. (2006) used data from the Las Cruces Trench site to explore the impact of a single-scaling-factor approach, i.e., Miller–Miller and Leverett scaling, and a multistep stochastic approach on solute transport. In contrast to the multistep stochastic approach, simulation results showed that both single factor scaling techniques, which explicitly assume that the shape factor of the soil moisture characteristic remains constant, seems to underestimate the heterogeneity at the experi-

mental site. Oliveira et al. (2006) further concluded, in accordance with Jury et al. (1987), that both the soil moisture characteristics and the hydraulic conductivity need to be measured to infer scaling factors for both functions separately.

Strengths and Limitations of Scaling

The attractiveness of the scaling approach resides in the fact that the observed variability of scaling factors may be captured in probability density distributions and in spatial-correlation structures. The spatial variability of all of the soil water processes is captured with one, two, or even three scaling variables (see above). Scaling in its various forms, particularly single-variable scaling, has therefore been a convenient approach to analyze the effect of spatially variable hydraulic conductivities on water flow (e.g., Peck et al., 1977; Sharma and Luxmoore, 1979; Hopmans and Stricker, 1989; Hopmans et al., 1988; Tseng and Jury, 1993; Roth, 1995; Rockhold et al., 1996; Birkholzer and Tsang, 1997; Kabat et al., 1997; Kim et al., 1997; Shouse and Mohanty, 1998; Wendroth et al., 1999; Deurer et al., 2001). The scaling method has proven its success not only in describing spatial variability of hydraulic properties in many field studies but also as a tool for numerical analyses.

One of the major limitations is the need for detailed information on the variability of soil hydraulic properties requiring extensive measurement campaigns using classical sampling techniques. Despite the substantial progress in soil physical measurement techniques to determine hydraulic functions, determination of all of the hydraulic functions for spatial variability analysis remains time consuming. Also the scaling approach typically requires more than one scaling factor (e.g., two or three—see, e.g., Jury et al., 1987) to describe the observed spatial variation in hydraulic properties. Although convenient, single-parameter scaling leads to significant underestimation of the true variability in the heterogeneous unsaturated flow field and it appears to be unsuitable for upscaling of unsaturated flow or transport processes (Oliveira et al., 2006). Fractal approaches are usually applied to characterize pore- to core-scale hydraulic properties or to quantify or characterize local-scale soil properties (Perrier et al., 2000; Perfect and Kay, 1991; Tyler and Wheatcraft, 1992; Pfeifer and Avnir, 1983; Van Damme and Ben Ohoud, 1990; Friesen and Mikula, 1987; Bartoli et al., 1999; Pachepsky et al., 2000), whereas the application of the fractal approach to analyze soil water processes at the field scale is still limited.

FORWARD UPSCALING APPROACHES

The Scaleway Approach

The scaleway approach is a general conceptual framework rather than a rigorous mathematical framework. In the scaleway approach, upscaling is implemented explicitly and in a stepwise approach across multiple scales (hence, the word *scaleway*). Spatial variability is considered to exist at multiple scales either continuously

(as in fractal systems) or in discrete steps (as in hierarchical systems). The scaleway approach breaks the system into multiple, discrete upscaling steps. At some arbitrary small step related to measurement scale, physicomathematical processes are defined explicitly and the process-relevant structure and topology of the vadose zone is defined exhaustively across the entire domain of interest. The effective properties and processes across that domain become the pixelized “small-scale” processes at the next higher level of upscaling. At the next higher level, a new set of measurements would be needed to define structure and topology at the level across a yet larger domain. The approach was recently presented by Vogel and Roth (2003) as a concept to handle multiscale heterogeneities and to predict flow and transport processes at a specific scale without making assumptions about the nature of the underlying structure because that structure is explicitly considered in this approach. This makes this approach rather flexible when compared with other upscaling approaches (e.g., stochastic perturbation analysis), which are typically limited to analyzing flow processes within specific spatial structures of the heterogeneous parameter field (e.g., joint-Gaussian) and only allow for upscaling across a finite change of scale.

The scaleway approach requires an explicit quantification of the structural properties of the soil at the smaller scale and avoids the need for effective process models at the scale of observation. The scaleway approach aims to bridge the gap between the structure of soils and their function (e.g., transport of solutes) by explicitly characterizing the relevant structural properties. The explicit consideration of the structural properties combined with the forward calculation of the flow process may, however, become numerically very demanding, especially when treating transient flow problems in three-dimensional flow domains. A few studies used high-resolution modeling of flow and transport processes in multi-Gaussian fields (e.g., Ababou and Gelhar, 1988; Desbarats, 1998) and have bridged the gap to stochastic approaches (see below). It is expected that these approaches may become more prevalent with increasing computational capabilities and with the improvement of numerical methods. The use of the scaleway approach is intimately linked to the use of powerful and novel numerical methods and models.

Up to now, this approach has been mainly applied to column-scale processes (Vogel, 1997, 2000; Vogel and Roth, 1998; Kasteel et al., 2000). At this scale, the flow process is typically described using physically based models such as Stokes equation, which is valid at the pore-scale level, or by the three-dimensional Richards equation (Eq. [14]) for variably saturated flow in continuum fields of hydraulic properties that are defined at scales larger than a pore but still smaller than a representative elementary volume. The flow process depends strongly on soil structure, which is determined by the three-dimensional pore geometry. At the pore-scale level, the geometry may be quantified in terms of pore size distributions described by various metrics and the topology of the structure referring to the way the struc-

tural units are connected in space. This connectivity can be described by, for example, the Euler characteristic, while the shape of the structural units may be quantified using Minkowski densities and functionals (Vogel, 2002). These Minkowski functionals are measures for quantifying arbitrary binary patterns such as those typically observed in porous media (Vogel et al., 2005; Roth et al., 2005). Similar approaches can be used to characterize the connectivity in continuous fields of hydraulic properties (e.g., permeability) at scales larger than a pore but still smaller than a representative elementary volume.

The need to specifically quantify the structure of the medium and the associated hydraulic parameters requires the use of tomographic methods such as x-ray tomography and image analysis techniques (Tarquis et al., 2003). As these methods do not determine hydraulic parameters directly, proxy relationships between the parameters of interest and the measured variables such as bulk density are needed. These proxy variables can then be used to distinguish structural units. One of the important questions in the scaleway approach is how to derive the $\psi(\theta)$ and $K(\psi)$ functions. At the core scale, this is solved by an explicit characterization of the pore geometry. Vogel (1997) determined the three-dimensional pore geometry of a soil core using serial sectioning of subsamples obtained from two different structural units. The hydraulic properties can then be estimated from two approaches. The first approach, which is numerically demanding, is based on the numerical solution of the Stokes equation in the full three-dimensional pore geometry. Computationally efficient solutions may be achieved using lattice Boltzmann simulation techniques (e.g., Krafczyk et al., 2000; Tölke et al., 2002). Vogel (1997) followed a second approach that reduces the original pore geometry by defining an equivalent network model consisting of idealized pores described by cylindrical tubes. The network model is generated using the pore size distribution and connectivity derived from the three-dimensional pore geometry of a soil sample. The $\psi(\theta)$ and $K(\psi)$ are then calculated for this equivalent network using Young–Laplace and Hagen–Poiseuille relationships, respectively. Vogel and Roth (1998) found good agreement between hydraulic properties obtained from this approach and those measured with multistep outflow. An interesting aspect is that in the scaleway approach, connectivity can be integrated in a direct way. Recent studies by Vogel (2000), Zinn and Harvey (2003), and Neuweiler and Cirpka (2005) have demonstrated the importance of accounting for connectivities of structural elements when upscaling water and solute processes, of upscaling from the pore scale to the continuum scale (e.g., Vogel, 2000), and of upscaling heterogeneities at the continuum scale (Zinn and Harvey, 2003; Neuweiler and Cirpka, 2005).

The strengths and limitations of the scaleway concept can only be evaluated in a preliminary fashion, as it has only recently been introduced and only few applications are presently available. It is conceptually very strong but the approach may be limited to small-scale systems. Especially the requirement to explicitly define the structure and the related hydraulic parameters at every point

in space may hamper its widespread use; however, the use of proxy variables seems to be an efficient way of circumventing this problem. Application at the field scale would require the use of “tomographic” methods to elucidate the underlying structure (see data fusion approaches, below). Especially the detection of highly connected structures is nearly impossible on the basis of grid observation points but would require a continuous mapping. The potential of hydrogeophysical methods (Vereecken et al., 2003; Rubin and Hubbard, 2005) such as spectral induced polarization (e.g., Titov et al., 2004), seismic methods (e.g., Rubin et al., 1992), ground penetrating radar, GPR (Lambot et al., 2004; Huisman et al., 2003) and electrical resistivity tomography, ERT (Kemna et al., 2003; Vanderborght et al., 2005) may offer the potential to characterize the spatial structure of soil hydraulic properties and flow and transport processes and should be further explored. Insofar as the scaleway concept heavily relies on indirect information about soil structure and topology, it cannot be strictly classified as a “forward upscaling” method, but is an important concept in the application of inverse modeling and especially the data fusion approach.

Stochastic Perturbation Approaches

Theoretical–Conceptual Overview

The scaleway approach requires a complete, deterministic knowledge of the small-scale structure to predict the large-scale behavior. Significant uncertainty typically exists, however, about the exact distribution of hydraulic properties in the unsaturated zone. We introduced geostatistical methods that can be used to statistically describe heterogeneous parameter fields from sparse data measurements. Using such (geo)statistical descriptors of a space variable as input, the so-called stochastic perturbation approaches allow us to statistically quantify upscaled properties, that is, quantify the mean state of the system at the larger scale and the local variance around the mean state at the larger scale, where the variance arises from the incomplete knowledge about the exact structure of the local-scale patterns of the vadose zone.

In the perturbation approach, it is assumed that the local flow $\mathbf{q}$ ($L T^{-1}$) is described by the Buckingham–Darcy equation:

$$\mathbf{q}(\mathbf{x}) = -K(\psi, \mathbf{x}) \nabla(\psi + z) \quad [13]$$

where z is the vertical coordinate, which is defined here to be negative in the downward direction, and $K(\psi, \mathbf{x})$ ($L T^{-1}$) is the hydraulic conductivity, which we assume here to be an isotropic soil property. In contrast to saturated flow, the hydraulic conductivity in an unsaturated medium is a function of the pressure head. The pressure head field is obtained by solving the Richards equation, which is a mass balance equation:

$$C(\psi, \mathbf{x}) \frac{\partial \psi}{\partial t} = \nabla[K(\psi, \mathbf{x}) \nabla(\psi + z)] + g(\mathbf{x}, \psi, t) \quad [14]$$

where $g(\mathbf{x}, t)$ is a sink–source term and $C(\psi, \mathbf{x})$ is the soil water capacity. When the parameters of the local-scale

hydraulic properties are treated as random space functions (see above), the flow (Eq. [13]) and mass balance (Eq. [14]) equations are stochastic continuum equations. The solution of the stochastic continuum equations is a stochastic quantity describing a probability distribution for each state variable rather than a deterministic quantity. Using the stochastic equation, moments of the probability distributions of the (dependent) state variables, for example, the water content, pressure potential, and water fluxes, can be computed. Solutions of the stochastic continuum equation often assume a Gardner–Russo parameterization of the hydraulic functions due to its linearity (e.g., Yeh et al., 1985a, 1985b, 1985c; Harter and Yeh, 1998; Russo et al., 1997) described by Eq. [7] and

$$\theta_e(\mathbf{x}, t) = (\theta_s - \theta_r) \{ \exp[0.5\alpha(\mathbf{x})\psi(\mathbf{x}, t)] \times [1 - 0.5\alpha(\mathbf{x})\psi(\mathbf{x}, t)] \}^{2/(m_g + 2)} \quad [15]$$

where θ_e is the effective moisture content and m_g is a parameter related to tortuosity. Solutions for other parameterizations have been presented as well. Mantoglou (1992) presented solutions for generalized hydraulic functions. Zhang and Winter (1998) used the Brooks–Corey model (1964) described as

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} = \left(\frac{\psi}{\psi_b} \right)^b = (\alpha\psi)^b \psi < \psi_b \quad [16]$$

$$K(S_e) = K_s(S_e)^{2b+3} \psi < \psi_b \quad [17]$$

$$S_e = 1; K = K_s \psi \geq \psi_b \quad [18]$$

where b is a parameter related to pore size distribution. Tartakovsky et al. (2003) illustrated that the predictability increases when Brooks–Corey or Mualem–van Genuchten parameterizations are used. The parameters α [or $\ln(\alpha)$] and $\ln(K_s)$ are assumed to be Gaussian random space functions, whereas θ_s , θ_r , and m_g are mostly assumed to be deterministic. A constant m_g implies that the hydraulic functions can be linearly scaled to reference functions (see scaling of hydraulic properties, above). The parameters and variables are expressed in terms of their mean or expected value and a perturbation:

$$\ln K_s = F + f \quad [19]$$

$$\alpha = A + a \quad [20]$$

$$\psi = H + h \quad [21]$$

$$C(\psi) = \Gamma(\psi) + \gamma(\psi) \quad [22]$$

$$q_i = Q_i + q'_i \quad [23]$$

where the uppercase characters represent the expected values and the lowercase or primed characters are the perturbations. The perturbations f and a are related to the scaling factors λ_K and λ_ψ as

$$\lambda_K = \exp(f) \quad [24]$$

$$\lambda_\psi = \frac{A}{A + a} \quad [25]$$

The hydraulic functions $K(\psi)$ and $\theta(\psi)$ are written in terms of the expected values and perturbations as:

$$K(\psi) = \exp[F + f + (A + a)(H + h)] \\ \approx \exp[F + AH](1 + f + Ah + ah) \quad [26]$$

$$\theta(\psi) \approx \theta(H) + [\Gamma(H) + \gamma(h)]h \quad [27]$$

A linearized perturbation equation (i.e., an equation that is linear in the perturbation terms) is obtained by: (i) substituting Eq. [26] and [27] with Eq. [21–23] into Eq. [13] and [14], (ii) neglecting all higher order perturbation terms, and (iii) subtracting the expected value of the equation. For instance, for steady-state flow and constant parameters A and F , the perturbation equation is (Zhang, 2002, Eq. [5.20], p. 226)

$$\nabla^2 h(\mathbf{x}) + A[2J(\mathbf{x}) - \nabla z(\mathbf{x})]^T \nabla h(\mathbf{x}) - A \frac{g(\mathbf{x}, H)}{K_m(H)} h(\mathbf{x}) \\ = -J^T(\mathbf{x})[\nabla f(\mathbf{x}) + H(\mathbf{x})\nabla a(\mathbf{x})] + \frac{g(\mathbf{x}, H)}{K_m(H)} f(\mathbf{x}) \\ + \left[\frac{g(\mathbf{x}, H)}{K_m(H)} H(\mathbf{x}) - J^T(\mathbf{x})\nabla H(\mathbf{x}) \right] a(\mathbf{x}) \quad [28]$$

where $K_m(H) = \exp[F + AH]$ and $J(\mathbf{x}) = \nabla[H(\mathbf{x}) + z(\mathbf{x})]$ is the mean hydraulic gradient and T refers to transposition. Moment equations are obtained by multiplying the linearized perturbation equation with a perturbation at a location $\mathbf{x}$ and taking the ensemble mean. Multiplying Eq. [28] by $h(\mathbf{x})$ and taking the ensemble mean yields

$$\nabla_x^2 C_{hh}(\mathbf{x}, \mathbf{x}) + A[2J(\mathbf{x}) - \nabla_x z(\mathbf{x})]^T \nabla_x C_{hh}(\mathbf{x}, \mathbf{x}) \\ - A \frac{g(\mathbf{x}, H)}{K_m(H)} C_{hh}(\mathbf{x}, \mathbf{x}) = -J^T(\mathbf{x})[\nabla_x C_{fh}(\mathbf{x}, \mathbf{x}) \\ + H(\mathbf{x})\nabla_x C_{ah}(\mathbf{x}, \mathbf{x})] + \frac{g(\mathbf{x}, H)}{K_m(H)} C_{fh}(\mathbf{x}, \mathbf{x}) \\ + \left[\frac{g(\mathbf{x}, H)}{K_m(H)} H(\mathbf{x}) - J^T(\mathbf{x})\nabla_x H(\mathbf{x}) \right] C_{ah}(\mathbf{x}, \mathbf{x}) \quad [29]$$

where $C_{uv}(\mathbf{x}, \mathbf{x})$ is the covariance of u at location $\mathbf{x}$ and v at location $\mathbf{x}$ and ∇_x is the gradient at location $\mathbf{x}$. To solve Eq. [29] for the pressure head covariances, the cross-covariances of the pressure heads and the hydraulic parameters, C_{fh} and C_{ah} must be derived by solving the following moment equations that are obtained by multiplying Eq. [28] (which is now evaluated at location $\mathbf{x}$) with $f(\mathbf{x})$ or $a(\mathbf{x})$ and taking the ensemble mean:

$$\nabla_x^2 C_{fh}(\mathbf{x}, \mathbf{x}) + A[2J(\mathbf{x}) - \nabla_x z(\mathbf{x})]^T \nabla_x C_{fh}(\mathbf{x}, \mathbf{x}) \\ - A \frac{g(\mathbf{x}, H)}{K_m(H)} C_{fh}(\mathbf{x}, \mathbf{x}) = -J^T(\mathbf{x})[\nabla_x C_{ff}(\mathbf{x}, \mathbf{x}) \\ + H(\mathbf{x})\nabla_x C_{fa}(\mathbf{x}, \mathbf{x})] + \frac{g(\mathbf{x}, H)}{K_m(H)} C_{ff}(\mathbf{x}, \mathbf{x}) \\ + \left[\frac{g(\mathbf{x}, H)}{K_m(H)} H(\mathbf{x}) - J^T(\mathbf{x})\nabla_x H(\mathbf{x}) \right] C_{fa}(\mathbf{x}, \mathbf{x}) \quad [30]$$

$$\nabla_x^2 C_{ah}(\mathbf{x}, \mathbf{x}) + A[2J(\mathbf{x}) - \nabla_x z(\mathbf{x})]^T \nabla_x C_{ah}(\mathbf{x}, \mathbf{x}) \\ - A \frac{g(\mathbf{x}, H)}{K_m(H)} C_{ah}(\mathbf{x}, \mathbf{x}) = -J^T(\mathbf{x})[\nabla_x C_{af}(\mathbf{x}, \mathbf{x}) \\ + H(\mathbf{x})\nabla_x C_{aa}(\mathbf{x}, \mathbf{x})] + \frac{g(\mathbf{x}, H)}{K_m(H)} C_{af}(\mathbf{x}, \mathbf{x}) \\ + \left[\frac{g(\mathbf{x}, H)}{K_m(H)} H(\mathbf{x}) - J^T(\mathbf{x})\nabla_x H(\mathbf{x}) \right] C_{aa}(\mathbf{x}, \mathbf{x}) \quad [31]$$

Using Eq. [29] to [31], the pressure head covariance functions can be obtained if the spatial (cross-)covariances of the hydraulic parameters are known and the boundary conditions are defined. Also, the expected pressure head H must be known to solve Eq. [29] to [31]. The value of H can be determined by solving the Richards equation (Eq. [14]) using the expected values of the hydraulic parameters, which is the so-called zeroth order approximation of H : $H = \psi^{(0)}$. The advantage of this approach is that H can be calculated directly and that the moment equations become a set of coupled linear equations. The disadvantage is that the effect of spatial variability of the hydraulic parameters on the mean or expected flow variables is not considered.

To describe the effect of the spatial variability of hydraulic properties on the mean flow process, higher than first-order perturbation terms are considered in the expansion of the flow equation (Eq. [13]). Neglecting higher than second-order perturbation terms, the expected value or the ensemble average of the water flux Q_i is

$$Q_i(\mathbf{x}) \approx -K_m \left(J_i \left\{ 1 + C_{ah}(\mathbf{x}, \mathbf{x}) + \frac{1}{2} [C_{ff}(\mathbf{x}, \mathbf{x}) \right. \right. \\ \left. \left. + A^2 C_{hh}(\mathbf{x}, \mathbf{x}) + H^2 C_{aa}(\mathbf{x}, \mathbf{x}) \right. \right. \\ \left. \left. + 2AC_{fh}(\mathbf{x}, \mathbf{x}) + 2HC_{fh}(\mathbf{x}, \mathbf{x}) + 2AHC_{ah}(\mathbf{x}, \mathbf{x}) \right] \right\} \\ \left. + E \left[(f + Ah + Ha) \frac{dh}{dx_i} \right] \right) \quad [32]$$

where E is the expected value. Using Eq. [32], an effective hydraulic conductivity $K_{\text{eff},ij}$ can be defined as

$$K_{\text{eff},ij} = -\frac{Q_i}{J_j} \quad [33]$$

It can be shown that when the coordinate axes align with the principal axes of the structural heterogeneity of the hydraulic parameters, the effective conductivity tensor becomes a diagonal matrix (i.e., $K_{\text{eff},ij} = 0$ for $i \neq j$). Using the expected or ensemble averaged fluxes in a mass balance equation, an upscaled flow equation, which describes the mean or expected water contents, fluxes, and pressure heads, is obtained:

$$\frac{\partial \Theta}{\partial t} = -\nabla Q \quad [34]$$

where Θ is the expected water content. In Eq. [34], local sinks or sources are not included but these could be implemented in a straightforward manner. The effective

soil water capacity C_{eff} can be defined and approximated as:

$$C_{\text{eff}}(H) = \frac{d\Theta}{dH} = \frac{dE[\theta(\psi)]}{dH} \approx \frac{dE[\theta(H)]}{dH} + \frac{dE(\gamma H)}{dH} = \Gamma(H) + \frac{dE(\gamma H)}{dH} \quad [35]$$

where $E(\gamma h)$ is the covariance between the soil water capacity and matric potential head perturbations. This term has been neglected in most cases since its effect on the large-scale flow is not as important as the effect of the variability of the hydraulic conductivity (Mantoglou and Gelhar, 1987b, 1987c).

Substituting Eq. [33] and [35] into Eq. [34] yields an upscaled flow equation of a similar form as the local flow equation:

$$C_{\text{eff}}(H) \frac{\partial H}{\partial t} = \nabla[K_{\text{eff}} \nabla(H + z)] \quad [36]$$

Main Results

Although the form of the upscaled flow equation is equal to the local one, the spatial variability of the local hydraulic properties leads to some important differences between the averaged or upscaled flow process and the local one. The variances of hydraulic parameters and the pressure head as well as their covariances in Eq. [32] and [33] show that the effective hydraulic conductivity is determined by the spatial variability of the local hydraulic parameters. The variance of the pressure heads and the covariances between pressure heads and hydraulic parameters, which are obtained by solving the moment equations (Eqs. [29–31] and the expected water flux Q_i , depend in a nonlinear way on the expected hydraulic gradient J . For transient flow conditions, they also depend on the change of pressure head with time. As a consequence, the effective hydraulic conductivity is a function of the hydraulic gradient and of its history, which implies a hysteretic behavior of the larger scale system.

Yeh et al. (1985a, 1985b, 1985c) studied steady-state unsaturated flow in soils with an isotropic structure and in stratified soils for vertical mean infiltration through an unbounded heterogeneous flow domain. The resulting stochastic flow equation was solved using spectral representation techniques. Two key results were obtained from the theoretical analysis. First, the variance of the mean capillary pressure increases monotonically when α and K_s (Eq. [7]) are independent. For a negative correlation between α and K_s , which represents a Miller–Miller similar medium, the variance of the mean pressure head is a non-monotonic function of the degree of saturation or the mean capillary pressure. Second, spatial heterogeneity of local hydraulic properties induces an anisotropy of the effective hydraulic conductivity. This is nicely illustrated by the work of Yeh et al. (1985c) using data from a sandy loam and a silty clay loam (Fig. 3). The anisotropy ratio of the effective unsaturated hydraulic conductivity calculated from stochastic perturbation theory depends strongly on the mean

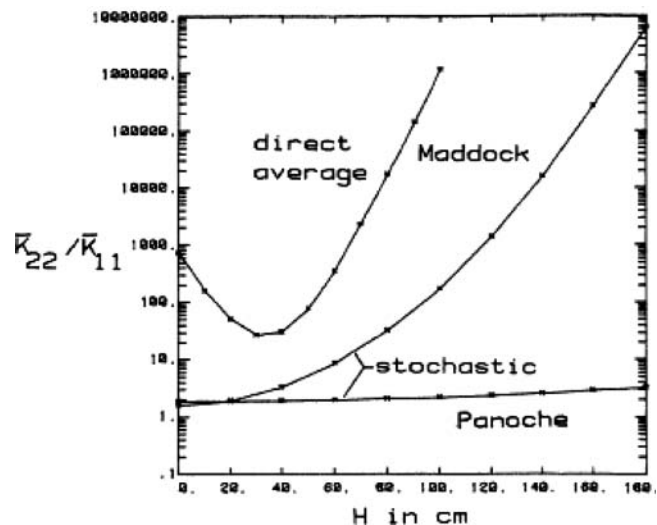


Fig. 3. Comparison of the anisotropic ratios obtained from the stochastic theory and the direct average methods for Maddock sandy loam and for Panoche silty clay loam; H is the mean capillary pressure head (from Yeh et al., 1985c; reproduced with permission of the American Geophysical Union).

capillary pressure and increases when the soil gets drier. The direct averaging method leads to a higher anisotropy than in the stochastic theory. This method takes the ratio of arithmetic and harmonic mean conductivity to estimate the effective conductivity anisotropy (for details, see Yeh et al., 1985a). In their derivation for the cases where $K_s(\mathbf{x})$ and $\alpha(\mathbf{x})$ are spatially variable, both parameters were assumed to have the same correlation scale and covariance function. They verified their theoretical findings using the field data of Nielsen et al. (1973) obtained on a 150-ha field site and various laboratory experiments. Yeh et al. (1985c) concluded that a major limitation in applying the stochastic method to upscaling lies in the determination of the statistical properties of $K_s(\mathbf{x})$ and $\alpha(\mathbf{x})$.

Yeh (1989) used a one-dimensional Monte Carlo analysis to verify the analytical results of his three-dimensional stochastic theory for steady-state flow. He confirmed the above findings with respect to the dependence of the variance of the pressure head on the mean pressure head. Moreover, the effective α parameter in the Gardner equation was found to depend on the mean pressure head. For the case of uncorrelated $\ln K_s$ and α , the effective value of α increased monotonically with increasing mean pressure head. In the case where both parameters were positively correlated, a critical mean pressure head existed corresponding to a minimum effective value of α . The dependence of α on the mean pressure head suggests that the effective hydraulic conductivity cannot be described by a Gardner exponential function. This indicates that functional forms typically chosen to describe local-scale hydraulic properties may not always be applicable to describe effective fluxes at the larger scale. The moisture- or suction-dependent behavior of the effective unsaturated hydraulic conductivity was verified in numerical simulations or through (quasi)analytical solutions of the stochastic flow equation (Wallach and Zaslavsky, 1991;

Polmann et al., 1991; Green and Freyberg, 1995; Khaleel et al., 2002).

Mantoglou and Gelhar (1987a, 1987b, 1987c) extended the analysis of Yeh et al. (1985a, 1985b) to transient unsaturated flow assuming that the hydraulic parameter fields are a realization of three-dimensional cross-correlated, second-order, stationary, statistically anisotropic, hydraulic parameter fields. Their analysis showed that the effective hydraulic conductivity (K_{eff}) and the effective differential moisture capacity (C_{eff}) depend on the mean values of $\ln K_s(\mathbf{x})$, $\alpha(\mathbf{x})$, and $C(\psi, \mathbf{x})$, the stochastic properties of the soil properties and the mean flow characteristics such as mean pressure head H , its gradients, and change with time, $\partial H/\partial t$. The dependence on the mean pressure head gradient and $\partial H/\partial t$ suggests hysteresis of the effective parameters, which was shown to increase with increasing heterogeneity of local-scale parameters. Also, the variances of the local pressure heads and fluxes were found to depend on the larger scale gradients and mean flow conditions (wetting vs. drying; see also Yeh et al., 1985a, 1985b). In addition they confirmed that the anisotropy of the effective hydraulic conductivity depends on the mean soil moisture content.

Harter and Zhang (1999) studied steady-state water flow in heterogeneous unsaturated media and examined the effect of variance $\ln K_s$, $\ln(\alpha)$, the mean soil water tension, the tortuosity m_g in the Gardner–Russo model (Eq. [7] and [15]), and soil layering on the coefficient of variation of soil water content. Soil layering was introduced by considering $\ln K_s$ to be anisotropic with smaller correlation length in the vertical than the horizontal direction. They showed that soil water content variability increases with soil water tension, as was also found by Yeh (1989) for the case of uncorrelated Gardner parameters. Under dry conditions, the soil water content was found to be very sensitive to the value of $\ln(\alpha)$, whereas it was found to be sensitive to the value of the tortuosity in the wet range. At low moisture content, small changes in the variability of $\ln(\alpha)$ with respect to $\sigma_{\ln K_s}$ can lead to substantial changes in soil moisture content variability expressed as a coefficient of variation.

The results obtained in the former studies are based on spectral methods that assume unbounded media in which the mean flow variables are constant or change slowly. This implies that the length scale of the flow domain has to be much larger than the scale of heterogeneity of the relevant hydraulic properties. Russo (1992) relaxed these conditions by proposing an up-scaling procedure that combined the stochastic approach of Yeh et al. (1985a) with the use of effective block properties (e.g., unsaturated hydraulic conductivity) having a scale comparable with the scale of the formation heterogeneity defined by, e.g., correlation functions. Mantoglou (1992) extended the three-dimensional stochastic perturbation approach to the study of transient nonstationary flow in bounded domains. Zhang and Winter (1998), Zhang (1999), and Zhang and Lu (2002) used numerical solutions of upscaled moment equations and derived the variance of the pressure heads and flow velocities for nonstationary and non-steady flow fields.

Experimental Verification of Stochastic Theories

There are only a few studies in the literature dealing with the experimental verification of stochastic theories for soil water processes. Important studies were conducted by McCord et al. (1991), Jensen and Mantoglou (1992), and Wendroth et al. (1999) at the field scale, and Wildenschild and Jensen (1999) and Ursino et al. (2001) at the laboratory scale. McCord et al. (1991) performed hillslope tracer experiments. They derived the anisotropy ratio from simultaneous measurements of the mean hydraulic gradient and the mean water flux, which was estimated from the path of the center of mass of an injected tracer plume. Their analysis illustrated that there was a considerable anisotropy of the hydraulic conductivity, which was caused by layered anisotropy. Simulations of water flow using a state-dependent anisotropy of the hydraulic conductivity reproduced the observed tracer movement the best. Ursino et al. (2001) performed unsaturated tracer experiments in a sand tank that was randomly filled with blocks of 5 by 5 by 0.5 cm of three different sand types that were rotated at an angle of 45° to the horizontal. The tank was uniformly irrigated at the top and tracer experiments were performed for three different flow rates. The deviation of the trajectory of the tracer plumes from the vertical axis increased with decreasing flow rate. This confirms, at least qualitatively, that the flow rate in a medium with a stochastic macroscopic structure depends on the hydraulic conductivity anisotropy.

Wildenschild and Jensen (1999) modeled the observed effective flow behavior in unsaturated well-defined heterogeneous sand material using a sand box packed with coarse and fine sand. Steady-state flow conditions were imposed. They compared hydraulic parameter estimates obtained from arithmetic and geometric means, an inverse procedure, and the perturbation theory of Mantoglou (1992) with measured effective retention and hydraulic conductivity parameters. They found that the inverse, geometric average, and stochastic approaches agree with steady-state measurements of hydraulic characteristics. Capillary suction simulations were found to agree with measured ones for all averaging approaches except the arithmetic mean. Transient outflow could be simulated with deterministic models and with simulations based on effective parameters derived from direct measurements. Therefore, the inverse approach may be considered as an alternative to the stochastic approach.

Jensen and Mantoglou (1992) applied the theory of Mantoglou (1992) to predict the mean behavior and the variance of the matrix potential and soil moisture content determined at the Jyndevad field site (100 by 50 m). A total of 24 sampling points for each of the three soil layers was used to derive statistical parameters of the hydraulic parameters. For each point, continuous moisture retention characteristic and unsaturated hydraulic conductivity were quantified. Stationarity and ergodicity were assumed for the lateral scale, which can be justified by the larger lateral extent of the field site compared with the vertical direction. Due to the limited extent of

the data set, only the mean and variance of $\ln K(H)$, $\alpha(H)$, $\theta(H)$, and $C(H)$ could be derived from the data. The covariance functions were assumed to follow an exponential model and their correlation lengths were treated as unknown variables (their effect being analyzed in a sensitivity study). They found that the stochastic three-dimensional model with effective parameters predicted the average system behavior better than using predictions with geometric mean averages of local-scale parameters. Also, the variance of the moisture content and pressure head were reasonably well estimated. They concluded that the stochastic approach provides a rational framework for modeling unsaturated flow processes in heterogeneous soils. It should be noted that the experimental work was not conducted on a bare field soil but on soil that was covered by short-cut grass. The process of root water uptake was considered in their analysis by expressing this process as an average large-scale sink term that is a function of mean quantities, without establishing a functional relation between large-scale sink and local-scale transpiration.

An interesting aspect of the study of Jensen and Mantoglou (1992) is the fact that they found an increase in head variances in periods of wetting. In their application of the stochastic theory to the field data of Jyndeved, it can be seen in Fig. 4 that during winter time (wetting), the calculated variation in pressure head is larger than the variation observed in the summer period. This contradicts the findings of Ferrante and Yeh (1999), who used a first-order stochastic model to examine the effect of uncertainty in both boundary conditions and heterogeneity on the mean pressure head and pressure head variances under drying and wetting flow condi-

tions. They found that the head variances at the drying front were always greater than the initial steady-state head variances. One of the reasons for this difference might be the effect of transpiration leading to the reduction of variability. Vegetation controls like transpiration may create or destroy spatial and temporal variability of soil moisture, pressure head, and their correlations (Rubin and Or, 1993; Katul et al., 1997; Albertson and Montaldo, 2003; Teuling and Troch, 2005). In addition climate controls may also affect the statistical properties of soil moisture, especially in humid regions like Denmark. For example, Fousseureau et al. (2000) developed a stochastic model for transient water flow in heterogeneous soils that takes into account not only the spatial variability of hydraulic properties and but also the spatiotemporal variability of the boundary flux condition. They showed that for humid climates, the statistics of the soil water content, Darcy flux, and pore water velocity were dominated by the boundary flux variability rather than by the effect of soil heterogeneity. The importance of soil heterogeneity increased in deeper soil layers. As shown also by Rubin and Or (1993), the moisture storage capacity of soils tends to attenuate the effect of the boundary flux variability with depth.

Wendroth et al. (1999) studied the spatiotemporal patterns of soil water pressure head in two surface horizons of a sandy loam soil and a heavy clay soil. They found that for the sandy loam soil and the heavy clay soil (up to 30 cm deep), a critical value of soil water pressure head existed at which flow processes were rather homogeneous, causing low variance of soil water pressure head. Above this critical value (drier soil) the variance of soil water pressure head increased again. Wendroth et al.

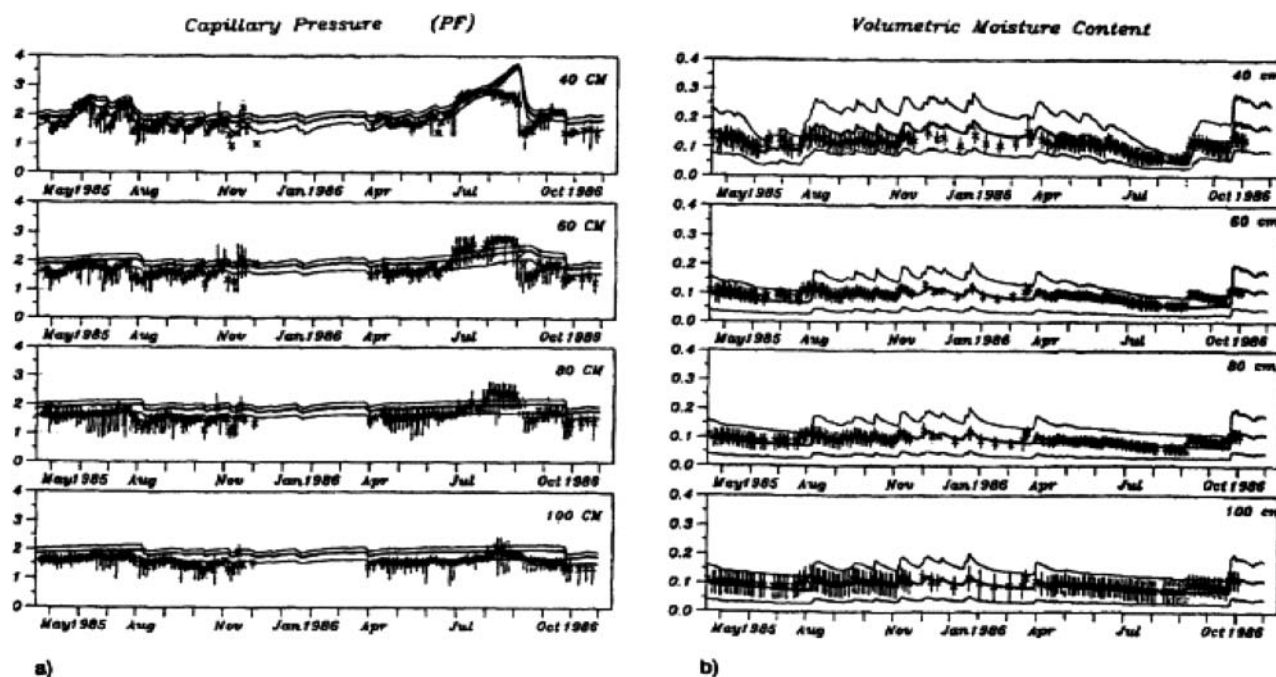


Fig. 4. Comparison of the means (thick curves) and variations (range of two standard deviations, thin curves) of (a) capillary tension ($P_f = \log \psi$, where ψ is pressure head) and (b) soil moisture content (θ) predicted by the stochastic model, with parameters semivariogram $\lambda = 5$ cm and correlogram $\rho = 2$, to measured values at the Jyndeved site. The crosses represent the mean, and the vertical bars indicate the range of variations of the local measurements (from Jensen and Mantoglou, 1992; reproduced by permission of the American Geophysical Union).

(1999) associated these observations with the finding of Roth (1995), who studied steady-state flow in a Miller-similar two-dimensional flow domain. In his study, Roth (1995) analyzed the statistical properties of the pressure head using the Mualem–van Genuchten model (van Genuchten, 1980) and Miller–Miller scaling to describe the variation in hydraulic properties. He found a critical pressure head value associated with a minimum variance. This variance increased as the soil became wetter or drier by imposing different flow conditions. These observations also correspond to the results obtained by Yeh (1989) using a stochastic perturbation and Monte Carlo analysis of steady-state flow in a heterogeneous flow medium. The critical pressure head with minimum variance can be shown to be caused by having negatively correlated K_s and α values (self-similar scaling). Even when α is log-normally distributed, as has been shown in field studies (e.g., White and Sully, 1992), perfect negative correlation between $\ln\alpha$ and $\ln K_s$ causes all unsaturated hydraulic conductivity curves to intersect near the critical head $h_{\min} = -\sigma_{\ln K_s}/(\alpha_g \sigma_{\ln \alpha})$, where α_g is the geometric mean of α , and $\sigma_{\ln K_s}$ and $\sigma_{\ln \alpha}$ are the standard deviations of $\ln K_s$ and $\ln \alpha$, respectively. As a result, the variance of the unsaturated hydraulic conductivity, $\ln K$, and of other unsaturated properties is smallest not at saturation, but when the mean head is at $h_{\min}$ (Harter, 1994, p. 156). Field observations of a critical pressure head with minimal pressure head variability at some state drier than saturation is therefore evidence of strongly scalable soil properties.

Strengths and Limitations of Three-Dimensional Stochastic Theories

Three-dimensional stochastic theories have provided fundamental insight in to the effect and importance of local-scale spatially variability of hydraulic parameters on effective soil water fluxes and parameters. These theories offer a consistent framework to define effective hydraulic parameters and even effective large-scale unsaturated flow equations (e.g., Mantoglou, 1992). A key result of these analyses, especially of the more recent higher order analyses and numerical Monte Carlo simulation studies, is that the upscaled governing equation has a similar form to the local equation. But the effective parameters in that equation are complexly dependent on the spatiotemporal patterns of the unsaturated system at the local scale. As a result, the upscaled water fluxes and moisture contents are not simply a function of the average parameter values obtained from measurements. Contributions to soil moisture and water fluxes instead often stem from higher order interactions between the patterns, distributions, and dynamics of time-invariant soil properties and time-variant, transient moisture dynamics, expressed in stochastic equations through nonlocal, nonstationary cross-covariances between system and state variables (Albertson and Montaldo, 2003; Harter and Hopmans, 2004).

There are significant constraints that limit the broad application of this approach. First, all perturbation approaches are only accurate for mildly variable soil water

conditions. These approaches do not directly apply to highly heterogeneous soils. Even in relatively homogeneous soils, the variability increases beyond the limits for which perturbation analysis is known to be accurate as the average value of the matric potential increases (drier soils).

The moment equations must be solved to determine the effective parameters of the upscaled flow equations from which the mean pressure heads and pressure head gradients are obtained. Since the mean pressure heads and pressure head gradients also appear in the moment equations, the moment equations and the upscaled flow equation are coupled. Mantoglou (1992) presented an iterative scheme for solution of the coupled flow and moment equations. But, we are not aware of attempts to solve these equations in a fully coupled way. Also, the coupled solution is not strictly consistent with a first-order perturbation approach since the terms containing the pressure heads and gradients, which are calculated using the effective flow model, are of higher order than the terms without pressure heads or gradients. To decouple the flow equation from the moment equation, Zhang and Lu (2002) use a zeroth order approximation of the pressure heads and gradients in the moment equations (i.e., the flow equation is solved using the expected rather than the effective hydraulic parameters). But the computational demand to solve the moment equations may be very high, especially for simulations of water flow under transient unsaturated conditions, even when the flow and moment equations are decoupled. For three-dimensional transient and nonstationary flow problems, it may be more efficient to implement a Monte Carlo simulation than to solve the moment equations. Monte Carlo simulations have frequently been used to numerically validate results obtained from three-dimensional stochastic theories (Yeh, 1989; Harter and Yeh, 1998) or to calculate ensemble-averaged quantities.

A major constraint in the application of three-dimensional stochastic theories resides in their high data requirement. Three-dimensional stochastic theories predicting effective hydraulic parameters, water fluxes, and state variables demand a detailed knowledge of the spatial structure of the hydraulic parameter fields. This has been pointed out in all studies addressing this topic and it is also the reason why results of stochastic theories are normally compared with numerical simulations rather than field measurements. Typically, the structure of these parameter fields (mean, variance, and correlation scale) has been derived from classical sampling approaches based on soil coring and local-scale *in situ* measurements. A significant need exists to develop novel measurement techniques for rapid assessment of spatial variability in unsaturated zone properties and for field validation of upscaling methods. It is our belief that this is one of the major challenges in vadose zone hydrology. Extensive field validation studies will contribute to an improved understanding about real-world soil systems.

A second constraint stems from the fact that the characterization of the variability in hydraulic parameter fields is based on multi-Gaussian distributions and statistics and provides the basis for analytical solutions

of effective parameters. The use of Gaussian statistics leads to patterns within which regions of extreme values (e.g., of high $\ln K_s$) are distinctly isolated from one another (limited connectivity). Zinn and Harvey (2003) found that flow and transport behavior in some porous-media flow systems that have log-normal univariate conductivity distributions and isotropic spatial covariance functions could be completely different from those in a truly Gaussian field due to specific patterns that allowed for connectivity between regions of extreme values. Neuweiler and Cirpka (2005) used homogenization theory (see other scaling approaches below) and stochastic averaging techniques to upscale the Richards equation in heterogeneous unsaturated porous media having such different connectivity properties. They compared permeability fields where highly permeable regions, intermediate-permeability regions, and low-permeability regions are well connected among themselves, but have the same statistics (mean, variance, and covariance) as a Gaussian random field. Their findings illustrate that the connectivity properties of the logarithmic intrinsic permeability field are important for the prediction of upscaled hydraulic parameters and effective water fluxes at the field scale. These connectivity structures are difficult to recover by classical local-scale sampling techniques.

A third constraint in the application of three-dimensional stochastic theories may be due to the fact that many vadose zone scientists are not familiar with the stochastic theories or lack the institutional and educational capacity to use these. Harter and Hopmans (2004) therefore suggested that “packaged” and well-documented stochastic software is needed to make these models more accessible to practitioners.

A fourth constraint pointed out by, e.g., Milly (1988) relates to the applicability of analytical solutions of the stochastic Richards equation to real field soils. The underlying assumption for the derivation of these solutions is based on the equivalence of spatial means and ensemble averages, assuming that the problem scale is much larger than the correlation distance of the hydraulic properties. For the vertical scale, which is typically shallow for soils (up to a few meters), this may constitute a serious limitation because correlation distances of hydraulic properties are typically not orders of magnitude smaller than soil thickness.

The Stream Tube Approach

In the stream tube approach, the horizontal heterogeneity in the vadose zone is conceptualized as a bundle of independent parallel soil columns, each with different but internally homogeneous soil hydraulic properties (Milly, 1988). Various constitutive equations defining one-dimensional flow within each soil column have been applied to this approach, including the one-dimensional Richards equation (see above) and the Green–Ampt approach (Smith and Diekkrüger, 1996; Rubin and Or, 1993).

The approach was first introduced by Peck et al. (1977) in their study of the effect of the soil spatial

variability on the terms of the water balance. The horizontal spatial variability of hydraulic properties was characterized using a geometric scaling factor α , with the assumption that its value is normally distributed. Based on evidence from field studies, Sharma and Luxmoore (1979) used a log-normally distributed scale factor in a similar analysis of the effect of spatial variability on soil water balance terms. Further studies based on the stream tube concept were performed by Smith and Hebbert (1979), Freeze (1980), Clapp et al. (1983), and Milly and Eagleson (1987). A fundamental and theoretical analysis of the effect of spatial variability on soil water processes using the concept of random space functions was given by Dagan and Bresler (1983) and Bresler and Dagan (1983). They analyzed the infiltration–redistribution process in a spatially heterogeneous saturated hydraulic conductivity field using the stream tube approach combined with the concept of piston flow. All other hydraulic parameters were assumed to be constant. This approximate description of the flow model was found to give accurate values of the expectations and variances of the field-scale flow variables. They also showed that effective hydraulic properties may be meaningful only under very restricted flow conditions, such as steady-state gravitational flow.

Water Flow and Effective Hydraulic Parameters

The stream tube approach has been used to analyze various concepts for estimating effective hydraulic parameters. Upscaled estimates of the Mualem–van Genuchten hydraulic parameters (K_s , α_{vg} , and n) have been computed using, for example, arithmetic mean values, geometric mean values, or static ensemble-averaged moisture retention characteristics and hydraulic conductivities (Smith and Diekkrüger, 1996; Zhu and Mohanty, 2002). These approaches have been evaluated against numerical simulations.

The static ensemble averaging of the Mualem–van Genuchten equations (Zhu and Mohanty, 2002) is defined as

$$\langle \Theta(h) \rangle = \left\langle \left[\frac{\theta_i - \theta_r}{\theta_s - \theta_r} \right] \right\rangle = \frac{1}{N_{st}} \sum_i^{N_{st}} [1 + (\alpha_{vg,i} \psi)^{n_i}]^{-1+1/n_i} \quad [37]$$

$$\langle K(\Theta) \rangle = \frac{1}{N_{st}} \sum_{i=1}^{N_{st}} K_{si} \frac{\{1 - (\alpha_i \psi)^{m_i} [1 + (\alpha_{vg,i} \psi)^{n_i}]^{-m_i}\}^2}{[1 + (\alpha_{vg,i} \psi)^{n_i}]^{m_i l}} \quad [38]$$

$$0 \leq \psi \leq \infty$$

where N_{st} is the number of stream tubes, l is a pore-connectivity parameter, and $m_i = 1 - 1/n_i$.

The ensemble-averaged hydraulic conductivity and moisture retention characteristics using mean values (arithmetic–geometric) are given by

$$\langle \Theta(h) \rangle = \Theta(\psi; \langle \alpha_{vg} \rangle, \langle n \rangle) \quad [39]$$

$$\langle K(h) \rangle = K(\psi; \langle \alpha_{vg} \rangle, \langle K_s \rangle) \quad [40]$$

The work of Smith and Diekkrüger (1996) considered random variation of hydraulic properties of a modified

Brooks–Corey relation. They performed calculations of water flow under steady-state and transient-state water flow using a stream tube approach and the Richards equation. Based on experimental data obtained in the Krummbach and Eisenbach catchments (Germany) and from a field experiment in Las Cruces, NM, they concluded that no significant correlation existed among any of the characteristic parameters. They therefore considered random variation in the hydraulic parameters to be independent. Field arithmetic mean values of K_s , θ_s , and θ_r can be used in flux simulations of heterogeneous fields without apparent bias on the dynamic behavior. Average parameters and static ensemble of, e.g., α_{vg} and n did not describe dynamic flow conditions (infiltration with redistribution). The inversely estimated effective parameters differed from the effective parameters obtained by static averaging and arithmetic averaging procedures. Here the effective parameters also appear to be dependent on the type of boundary condition. They concluded that there seems to be no simple way to predict field effective hydraulic parameters based on the local measurement of hydraulic parameters or boundary conditions.

Zhu and Mohanty (2002, 2003) used the stream tube approach to investigate the effect of parameter averaging schemes and their appropriateness in describing the ensemble field-scale behavior of heterogeneous soils. They considered steady-state vertical flow relating capillary pressure head and elevation above a water table. The generated parameter fields for the Gardner–Russo model (K_s and α) and the Mualem–van Genuchten model (K_s , α_{vg} , and n) were assumed to exhibit exponentially decaying covariance functions and to be cross-correlated. Hydraulic properties were assumed to be constant in the vertical direction. Random fields of 50 by 50 grid cells were generated with the domain length being equal to 10 correlation lengths. Three different averaging schemes were studied: the static ensemble averaging, the arithmetic mean values, and the geometric mean values of the parameters. They restricted their analyses to steady-state conditions of both infiltration and evaporation for given pressure heads at the boundaries. The effective parameters were considered to characterize an equivalent homogeneous medium that will discharge approximately the same amount of ensemble flux as the heterogeneous flow domain. The effective hydraulic conductivity parameters were found to be dependent on the type and nature of the boundary conditions, consistent with Smith and Diekkrüger (1996).

Especially for the case of coarse-textured fields, effective parameters were found to be very sensitive to surface pressure conditions. The accuracy of the average parameter approaches relative to the true upscaling approach was found to be dependent on the degree of correlation between the parameters. Under conditions of perfect correlations between parameters, average parameter approaches were very efficient and accurate when predicting steady-state, upscaled evaporation and infiltration fluxes. The spatial variability in α_{vg} had a larger impact on the ensemble behavior of soils than the variability in n , partly due to the fact that n can be determined with greater certainty. In case of a predominant

evaporation regime or infiltration conditions, the use of a geometric mean value for α_{vg} simulated the ensemble behavior better than an arithmetic average of α_{vg} .

Different conclusions were found with respect to the importance of horizontal flow when comparing both the stream tube and three-dimensional stochastic approaches. In the case of the stream tube approach, Smith and Diekkrüger (1996) and Zhu and Mohanty (2002) analyzed the magnitude of the horizontal flow components that might be due to the pressure differential between alternate cells and found them to be of minor relevance. Harter and Hopmans (2004), however, concluded that potentially significant lateral flow may occur in heterogeneous soils even under gravity-flow conditions, as found, for example, in the field experiment described in McCord et al. (1991), which was shown to be consistent with stochastic theory. Or and Rubin (1993) investigated the ability of a stream tube approach to reconstruct two-dimensional flow solutions obtained from the two-dimensional Richards equation for a heterogeneous flow domain under steady-state flow conditions. Monte Carlo simulations were used for the two approaches and compared with analytical solutions of the considered flow problem. They found that the vertical distributions of the average pressure head and the average saturation agreed well for all three approaches; however, the variance of the pressure head was found to be lower for the two-dimensional case compared with the stream tube approach, suggesting that in the two-dimensional approach, streamlines may bypass zones of low conductivity. A two-dimensional setting therefore leads to a larger dissipation of horizontal gradients in pressure head. This dissipation was found to be accompanied by an increased variability in the saturation due to increased discontinuities in the soil moisture profile. The latter were caused by the imposed smoothing of the pressure head. Under transient conditions, the spatial moments of pressure head and saturation could not be predicted using the stream tube approach.

Strengths and Limitations of the Stream Tube Approach

The stream tube approach allows a large variety of modeling concepts to be included in the analysis of soil water processes in heterogeneous fields. In addition, computational power continues to expand so that transient solutions of Richards equation are possible. The availability of satellite data to characterize the upper boundary conditions as well as the vegetation cover allows for a regional cover of soil water processes. The stream tube model approach can be applied to describe water flow processes at scales where lateral water redistribution in soils is not important. It assumes that the spatial structure of the parameter distribution in the lateral direction is not important or relevant. But this scale is much larger than the scale at which hydraulic properties and their variability are determined (the soil core scale). Hence, one fundamental problem is that we have no methods to determine the hydraulic properties and their spatial variability at such large scales. The second fundamental problem is that the application of a

stream tube model in fact a priori excludes the use of Richards' equation to describe the larger scale water flow. The Richards model assumes that the average water flow is a unique function of the gradient of the average energy status of the water, which results in a diffusive water flux. This assumption is appropriate if local gradients are not correlated with the local hydraulic properties. In contrast, the stream tube model assumes that local water flow is exclusively determined by the stream tube hydraulic properties. As a consequence, the large-scale averaged water flux predicted by a stream tube model is not diffusive and by definition inconsistent with Richards' equation as a large-scale model.

The Aggregation Concept

The concept of aggregation refers to estimating aggregated hydraulic characteristics that describe soil water processes in soil map units or numerical grids in soil-plant-atmosphere models and climate models that typically consist of a mixture of soil types. A soil type refers to a classification unit based on taxonomic criteria, while a soil map unit or a grid usually represents an association of related soil types, each with a specific areal proportion. Two approaches have been pursued in assigning aggregated hydraulic properties: aggregation based on defining an average textural composition combined with regression analyses to estimate hydraulic parameters, and estimation of the hydraulic properties based on the dominant textural fraction of the map unit or grid under consideration. The derivation of hydraulic properties in the aggregation approach is closely related to the use of pedotransfer functions (PTFs) that relate soil hydraulic properties to simple soil properties such as texture, bulk density, and soil organic matter (e.g., Bouma, 1989; Vereecken et al., 1989, 1990; Pachepsky and Rawls, 2004). See the respective literature for more details on the development and use of PTFs.

Noilhan and Lacarrère (1995) derived aggregated soil hydraulic parameters for an area corresponding to a typical grid size in global circulation models. They compared the calculated area-averaged fluxes based on three-dimensional model calculations, honoring the subgrid variability with a one-dimensional calculation using aggregated parameters. They found that the area-averaged fluxes calculated by the one-dimensional model agreed to within 10% with fluxes calculated by the three-dimensional model. Kabat et al. (1997) performed a simulation experiment using the aggregation procedure introduced by Noilhan and Lacarrère (1995) using the regression equation derived by Cosby et al. (1984) to further detail these results. They performed a simulation experiment using a surface grid composed of the textural classes clay, sand, and loam with fractional areas of 4/9, 3/9, and 2/9, respectively. Values for the parameters in Eqs. [16–18] were obtained from Cosby et al. (1984) for each of the textural classes. The area-aggregated texture was then estimated from the respective clay and sand fractions as a weighted arithmetic average and used to predict the Brooks–Corey b parameter and θ_s . Aggregated K_s values were derived from linearly averaging K_s and by estimating

K_s from the nonlinear relation between aggregated clay and K_s as derived by Cosby et al. (1984). Results were also compared with an aggregation approach based on the dominant textural class (clay). The use of the dominant textural class led to substantial underestimation of area-averaged evaporation and area-averaged drainage. The deep percolation flux was found to differ widely between the three textural classes, indicating that an aggregation approach based on the dominant soil type would also result in an underestimation of the area-averaged runoff. In addition, they found that the aggregated parameters may be used to predict the area-averaged evaporation flux. The aggregated parameters, however, failed to predict water balance terms associated with other soil water flow processes, such as downward percolation fluxes.

Regarding strengths and limitations, the aggregation approach is relatively straightforward and easy to apply, especially for large-scale problems. The development of this approach is mainly driven by the availability of soil data. The analysis of Kabat et al. (1997) showed that limitations exist in describing effective soil water processes like downward percolation fluxes. The aggregation approach is similar to the stream tube approach presented by Zhu and Mohanty (2002) with respect to the estimation of hydraulic properties. Both approaches aim to estimate the hydraulic properties from simple soil properties. In case of the aggregation approach, the multilinear regression equations of Cosby et al. (1984) were used, whereas Zhu and Mohanty (2002) used PTFs that do not rely only on textural properties. Also, the stream tube approach and the aggregation approach provide the same flexibility in describing soil water processes. Process description may range from simple capacity models to transient one-dimensional simulations using Richards equation. Thus, the results obtained from the stream tube approach, such as the dependency of effective parameters on boundary conditions or state variables, are expected to be also valid when using the aggregation approach to derive effective parameters. A significant shortcoming of the approach—intrinsic to the linear averaging processes used—appears to be its unreliable prediction of flux terms in the unsaturated system (evaporation, percolation, and root water uptake). Effectively, the approach amounts to a first-order approximation of simple stochastic models. It therefore ignores the effective contributions (positive or negative) to soil water fluxes from parameter and process interactions (in space and time) in a heterogeneous system that can only be captured by dynamic upscaling (e.g., higher order stochastic methods).

Other Upscaling Approaches

Further upscaling methods presented in the literature and of significant relevance to vadose zone flow processes include the volume averaging method, renormalization methods, homogenization methods, and percolation theories. Volume averaging methods (e.g., Whitaker, 1986; Quintard and Whitaker 1988, 1990a, 1990b; Whitaker, 1999) have been typically applied in upscaling microscopic two-phase flow processes to macroscopic field

variables and hydraulic conductivity functions. This approach was expanded to predict large-scale properties, for example, permeability tensors and large-scale capillary pressure-saturation relations based on large-scale averaging methods (e.g., Ahmadi and Quintard, 1996; Quintard and Whitaker, 1988), but no literature is presently available concerning the application of these methods to water flow in the vadose zone.

Applications of renormalization methods to water flow processes mainly address the calculation of effective or equivalent hydraulic conductivities in saturated porous media (e.g., King, 1987, 1989, 1993, 1996; King and Neuweiler, 2002; Hristopulos and Christakos, 1999) as well as the derivation of effective transport parameters (e.g., Jaekel and Vereecken, 1997; Hristopulos, 2003). Homogenization methods have only recently been applied to unsaturated flow problems at the laboratory (local) scale (Hornung, 1996; Aurialt, 2002; Lewandowska and Laurent, 2001; Lewandowska et al., 2004) and are restricted to specific flow regimes and heterogeneity structures (e.g., Neuweiler and Cirpka, 2005). Earlier work on homogenization mainly focused on upscaling single or two-phase flow from the microscopic to the local scale (e.g., Bourgeat, 1986; Arbogast, 1990) and two-phase flow in dual-porosity porous media (e.g., Bourgeat, 1984; Arbogast, 1993a, 1993b; Noetinger et al., 2001). To our knowledge, no closed-form expressions of effective hydraulic conductivity have been derived using homogenization theory for spatially variable hydraulic properties that would allow a direct comparison between, for example, stochastic perturbation theories and homogenization methods. The same is valid for the large-scale averaging methods. Percolation network approaches originally were used to derive macroscopic constitutive relationships for two-phase flow from an explicit consideration of pore-scale structure (e.g., Ferrand and Celia, 1992; Rajaram et al., 1997).

Another upscaling approach that is based on fundamental physics is the use of percolation theory. Hunt (2005) provided a comprehensive physical treatise on the application of percolation theory and critical path analysis to derive local- and large-scale effective properties. Large-scale properties are derived from the geometric patterns (topology) of the porous media, which in turn are analyzed for their connectivity patterns using concepts from percolation theory including the so-called percolation threshold, percolation cluster statistics, and percolation correlation length. The approach analyzes patterns of connectivity, clustering, and pore-throat distribution that generate certain statistics, which serve as the basis for computing effective (upscaled) properties. Upscaling is achieved across entire scaleways and is not limited to systems with finite variance (single-scale upscaling).

THE INVERSE UPSCALING APPROACHES

Recently, inverse modeling with many distributed data collected at various scales has been applied to obtain large-scale effective parameters. In contrast to the previous approaches, inverse methods start from infor-

mation, measurements, and observations of state variables and fluxes (see Fig. 1) to derive effective parameters of models that describe the system's behavior at a certain scale. Upscaling is achieved by (i) using methods that observe the system at a larger scale (e.g., remote sensing from satellite platforms), (ii) using smaller scale data as calibration targets for a system described (and defined) at a larger scale (e.g., water level measurements, average soil water tension, or soil water contents), or (iii) using spatial statistics and structures of local-scale state variables to infer the spatial structure of smaller scale process parameters, which can subsequently be used in forward upscaling.

Approaches used in (i) and (ii) are based on the assumption that the effective flow equations are known at the relevant (larger) scale. Inverse upscaling approaches do not lead to "upscaled" equations and in this sense they are different from, e.g., stochastic perturbation theories or volume averaging methods (see above). Effective equations obtained from forward upscaling could be implemented in inverse modeling. The upscaled form of the Richards equation as derived by Mantoglou (1992) implies that the effective unsaturated hydraulic conductivity depends on the state of the system (e.g., fluxes), which can hypothetically be implemented in an inverse modeling scheme. In approach (iii), information about the structure and variability of local-scale state variables and fluxes is used in inverse modeling to infer information about the structure of local parameter fields. This is generally done in a stochastic and geostatistical framework. This framework also offers the possibility to combine information or prior knowledge about parameter distributions with information about local state variables or other information that is correlated with state variables or parameters. We refer to the combined use of different information sources as *data fusion*.

First, we discuss the use of large-scale information, generally obtained from remote sensing techniques, to derive large-scale model parameters [approach (i)]. Subsequently, we discuss the use of local-scale measurements to derive model parameters of regions with homogeneous parameter distributions of which the extent is a priori known [approach (ii)]. Third, we discuss stochastic or geostatistically based methods to derive the structure of local parameter fields from local measurements [approach (iii)]. In a last part, we discuss the fusion of different data types to assess local parameter fields as a means to model at the larger scale.

Estimation of Large-Scale Model Parameters from Large-Scale Observations

Using satellite-based remote sensors, the surface status of large regions can be monitored. They provide spatially continuous information that is typically limited to the upper centimeters of the soil. Most applications are found at the catchment and regional scale, with a specific emphasis on characterizing soil moisture variability (e.g., Jackson and LeVine, 1996; Famiglietti et al., 1999; Grayson and Blöschl, 2000; Schmugge et al., 2002; Oldak et al., 2002a, 2002b; Wilson et al., 2003; Romshoo,

2004; Jacobs et al., 2004) and soil water fluxes or vegetation parameters (Noilhan and Lacarrère, 1995; Feddes et al., 1993; Burke et al., 1997; Wood, 1998; Boegh et al., 2004; Jhorar et al., 2002, 2004; Vrugt et al., 2004; Demarty et al., 2005; Schoups et al., 2005) often in combination soil-vegetation-atmosphere (SVAT) models. Remote sensing data have mainly been used in a descriptive manner or as input in the SVAT models but have not been linked to available stochastic theories (see above) to derive, e.g., effective parameters and soil water fluxes (e.g., root water uptake). Rather, scaling techniques have been developed to estimate, e.g., upscaled moisture contents (e.g., Peters-Lidard et al., 2001; Wood, 1997). Especially ground-based remote sensing may contribute to a better characterization of the spatial and temporal variation of moisture content and water and energy fluxes at the field scale (e.g., Albertson and Montaldo, 2003). Remote sensing techniques that show a clear potential in this respect are infrared sensing techniques and radar techniques, both passive and active ones (Hollenbeck et al., 1996; Schmugge et al., 1994; Wigneron et al., 1999; Schneeberger et al., 2004).

Using measurements representing a large scale to model a large-scale system is consistent with the classic inverse modeling approach. Within this framework, inverse modeling is a powerful method to derive large-scale (“effective”) hydraulic parameters as well as effective state variables and fluxes. The scientific basis of the approach, including its strengths and weaknesses, is well established and has been used in many fields of research. Recently, Aronica et al. (1998) and Beven (2006) addressed the issue of equifinality in hydrology, which is directly related to the uncertainty contained in inversely estimated parameters. Inverse modeling typically uses temporally and spatially distributed information on state variables of the system being studied that are relatively inexpensive to measure. Direct information concerning the spatial statistical structure of hydraulic parameters is not required but may be helpful in constraining the inverse estimation procedure. A limitation of most of these methods is that an a priori assumption must be made about the large-scale process function. For the vadose zone, the large-scale process is typically assumed to be conceptually identical to that controlling the small-scale process (e.g., Richards equation). At present, however, there is no rigorous evidence that the Richards equation, assumed to be valid at the local scale, is also valid at the field or even larger scale.

Estimation of Effective Parameters of Larger Scale Homogeneous Regions with Known Spatial Extent from Local Measurements

One method that assumes a single homogeneous region of larger scale effective properties is the spatial moment analysis of local state variables. It is applicable to transient infiltration experiments from point or line sources. The spatial moment analysis exploits the information contained in the temporal evolution of the spatial distribution of system variables such as soil moisture content or solute concentration from the point or line

source. The average spreading rate (captured quantitatively by computing spatial moments and plotting those as functions of time) is inversely fitted using a homogeneous, large-scale model to obtain effective or “upscaled” hydraulic or transport parameters. Mostly used for contaminant transport analysis, there are presently few studies that use soil moisture or soil water tension data to estimate effective hydraulic properties from spatial moments. Yeh et al. (2005) and Ye et al. (2005) used spatial moment analysis to derive three-dimensional effective unsaturated hydraulic conductivity tensor from snapshots of three-dimensional moisture distributions under transient-flow conditions. Application of this approach to data from the Hanford site showed that the effective hydraulic conductivity exhibited a moisture-dependent anisotropy. The principal directions of the spatial moments were found to be depth dependent as the plume evolved through the subsurface. The effective hydraulic conductivities compared well with laboratory-scale measurements of unsaturated hydraulic conductivity. Ward et al. (2006) also used spatial moment analysis to derive the upscaled hydraulic conductivity and also found moisture-dependent anisotropy. They concluded that this approach could capture the mean plume behavior of soil moisture but not the asymmetry caused by heterogeneity and lateral spreading.

Another approach was proposed by Zhang et al. (2003, 2004). They combined parameter scaling and an inverse technique to upscale hydraulic parameters from the local to the field scale using a two-step approach. In this method, the soil is considered to be heterogeneous and treated as a composition of multiple equivalent homogeneous media (EHM) whose hydraulic properties are scale dependent. The proposed parameter scaling technique relates the mean parameters in the hydraulic functions (e.g., K_s , α_{vg} , and n in the Mualem–van Genuchten equations) derived from local-scale measurements from a selected EHM to effective parameter values $K_{s,e}$, n_e , and $\alpha_{vg,e}$ of this EHM. The general relationship for any of the hydraulic parameters in, e.g., the Mualem–van Genuchten, can be written as

$$\beta_{e,i,j} = C_{i,j} \mu_{i,j} \quad [41]$$

where $\beta_{e,i,j}$ is the effective value of the i th parameter for the j th EHM and $\mu_{i,j}$ is the mean value derived from local-scale measurements. A simplifying assumption, due to the fact that the spatial variability of each hydraulic parameters is not readily available or known, postulates that the coefficients $C_{i,j}$ are the same for all the EHM. Selecting one of the EHM as a reference EHM and using the simplifying assumption, the following scaling relations are obtained:

$$\beta_{e,i,j} = \frac{\mu_{i,j}}{\tilde{\mu}_i} \tilde{\beta}_i \quad [42]$$

where $\tilde{\beta}_i$ is the effective parameter of the reference EHM for the i th parameter and $\tilde{\mu}_i$ is the mean parameter value of the reference EHM. In the second step, the parameters of the reference EHM are determined by minimizing the sum of weighted squared residuals between calculated and observed state variables like

moisture content or pressure head obtained from, e.g., well-designed infiltration experiments. Calculations can typically be done using Richards' equation (Zhang et al., 2004). The approach of Zhang et al. (2004) relies strongly on the knowledge of the structural organization of the EHM. For applications in upscaling soil hydraulic properties, this information needs to be obtained from local drilling information or soil maps. In addition, prior knowledge of the parameter values can be included in the minimization. Ward et al. (2006) applied this approach, also referred to as *effective medium approximation*, to derive the upscaled water retention characteristic and unsaturated hydraulic conductivity from the observed moisture distribution in the frame of a field-scale water injection test. They found that the up-scaled $K(\theta)$ showed evidence of saturation-dependent anisotropy, a finding also obtained using forward upscaling methods.

Finally, the approach presented by Eching et al. (1994) uses the scaled Richards equation (above) to estimate effective hydraulic properties at the field scale. They inversely estimated field-representative hydraulic functions using rescaled measured cumulative drainage curves.

Application of spatial moment analysis provides a direct means to calculate effective hydraulic conductivity. The use of moisture content is attractive as it is a state variable that is comparatively inexpensive to determine (e.g., time domain reflectometry [TDR] measurements). But the high spatial resolution required may limit its application. In the future, the use of hydrogeophysical and ground-based remote sensing methods may definitely provide high-quality data sets for deriving effective hydraulic parameters with this method. Another problem is that the spatial structure or spatial arrangement of the larger scale homogeneous units must be defined a priori and this information is often not available or is uncertain.

Inverse Modeling of Local Parameter Distributions from Local Observations of State Variables

In this approach, the spatial distribution of the parameter field is derived from local observation of state variables and fluxes. In general, the size of the parameter vector that needs to be estimated is much larger than the size of the data vector. Hence, the inversion problem is underdetermined. The inversion problem must therefore be regularized or conditioned on other a priori information. This type of inverse modeling has been developed and applied for some time in groundwater hydrology (e.g., Kitanidis and Vomvoris, 1983; Hoeksema and Kitanidis, 1984; Hanna and Yeh, 1998). Overviews of different methods or procedures used in the inverse groundwater problem are given by McLaughlin and Townley (1996) and Zimmerman et al. (1998).

The inclusion of site-specific, local-scale information on system variables (model parameters, e.g., saturated hydraulic conductivity) or state variables (e.g., soil water tension) can be achieved by conditioning the geostatistical, local-scale distributions of the system parameters (variables) on local knowledge (usually point measure-

ments) of system or state variables (Harter and Yeh, 1996; Yeh and Zhang, 1996). Harter and Yeh (1996) proposed a conditional stochastic modeling implementation of this approach to investigate the value of local-scale tensiometer data (soil water tension) in upscaling water and solute transport in the vadose zone. Their approach combines the cokriging method, a linearized approximation of the soil water tension first-order perturbation solution, and a finite element model solving Richards' equation within a Monte Carlo simulation framework. Local-scale information and its use in this type of inverse upscaling was shown to be important only when local-scale data existed in a relatively dense network, that is, with measurements at intervals not much larger than the spatial correlation scales of soil water tension. The approach of Harter and Yeh (1996) was expanded by Yeh and Zhang (1996) to also account for nonuniform and nonstationary flow problems under field conditions.

The conditional stochastic modeling approach is a hybrid forward and inverse modeling approach. It is a forward upscaling approach in that it applies a priori knowledge of the geostatistical distribution of unsaturated hydraulic parameters to yield upscaled values of, for example, unsaturated flow. It is an inverse method in that it also accounts for specific, location-dependent, local-scale outcomes of state variables (e.g., moisture content) as part of the upscaling approach. As a result, in contrast to unconditional stochastic methods, the moments of both system and state variables in this approach are therefore intrinsically nonstationary. For example, at the measurement location of a state variable, its local mean is equal to the measured value and its variance is equal to the measurement error. The spatial distribution of the stochastic moments of the state variable is also affected by knowledge of the local values of system variables. The conditional stochastic (upscaling) approach is therefore most useful where a large amount of local-scale data exist. In addition, the use of first-order perturbation theory to condition the Monte Carlo simulation implies that the conditioning moments are strictly valid only for mildly heterogeneous soil systems (see also above). The constraining effect of the conditioning data, however, allows application of this approach also to strongly heterogeneous systems. The approach was later expanded to include geophysical information and became the gateway for the development of data fusion techniques (Yeh et al., 2002).

The Data Fusion Approach

Data fusion is an expansion of the stochastic conditioning approach. In this approach, any information that can statistically be (cor)related to either state or system variables of interest or to boundary fluxes is applied to constrain the inversion problem. Data fusion basically uses explicit structural information about hydraulic properties combined with information about the spatial and temporal distribution of state variables as well as all other relevant a priori knowledge. Stochastic data fusion methods recently discussed in the literature

provide a rational framework that allows exploitation of all relevant data and a priori knowledge (e.g., Yeh and Simunek, 2002; Yeh et al., 2002; Liu and Yeh, 2004; Kowalsky et al., 2004, 2005).

Yeh and Simunek (2002) proposed a stochastic fusion concept to characterize and monitor vadose zone processes. They proposed a geostatistically based inverse approach as an appropriate conceptualization of the inverse problem combined with an assimilation of different types of information. Their stochastic fusion approach comprises two levels. At the first level, the hydrological or geophysical “soft” information is used to independently improve the spatial characteristics of the primary variables. Hydrological data may include hydraulic conductivity measurements at various locations, including its spatial characteristics, moisture content, and pressure head data in space and time. These can be obtained either from direct measurements on soil cores or from monitoring experiments using, for example, TDR or GPR. The electromagnetic properties may include point measurements of resistivity and dielectric constants and parameters of the relationships among moisture content, resistivity, and dielectric constants. Yeh and Simunek (2002) proposed using a sequential linear estimator to obtain conditional means and covariances of the state variables and parameters for three-dimensional, steady-state flow conditions. This corresponds to the first of two stages in the stochastic fusion concept. The second stage of the stochastic fusion concept is depicted in Fig. 5 and shows the various steps for the case of character-

izing soil moisture dynamics and electromagnetic properties of the vadose zone. Rather than limiting the concept to ERT, we included other geophysical techniques such as cross-borehole GPR that may provide information about the soil moisture dynamics. In principle, other geophysical methods such as seismic techniques or surface nuclear magnetic resonance methods may be used to characterize soil moisture dynamics (Yaramanci et al., 2005). Results from ERT and GPR surveys and from cross-borehole measurements may be used to derive functional relationships between resistivity, the dielectric constant, and moisture content and to obtain estimates of mean values and covariances of resistivity, moisture content, the dielectric constant, and hydraulic parameters. Joint geophysical inversion may provide information on the conditional means and covariances of moisture content, resistivity, and dielectric constants. The information on spatial and temporal variation in soil moisture may then be used in inverse modeling of soil water dynamics to obtain estimates of hydraulic parameters, including their statistical properties. The improved estimates of soil moisture content and its statistical properties are then returned to the geophysical inversion module. This iteration is continued until no further improvement is obtained in hydraulic parameters, moisture content, or geophysical quantities. Yeh and Simunek (2002) provided various examples of this general flow scheme.

Kowalsky et al. (2004) developed an inverse technique that allows estimation of flow parameter distribu-

Stochastic Data Fusion

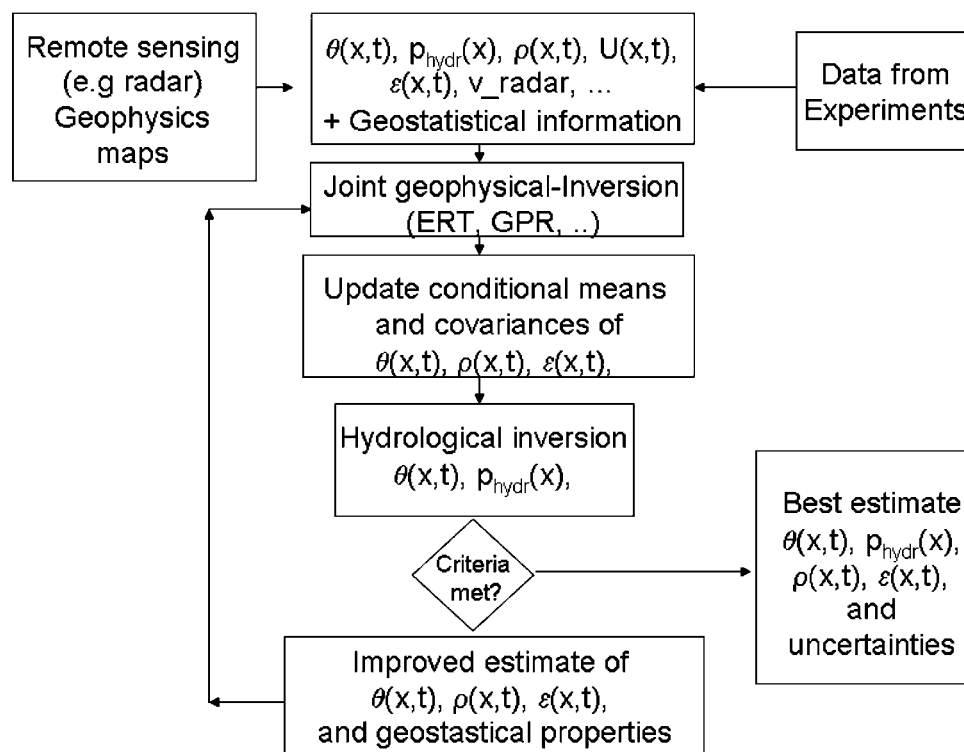


Fig. 5. Stochastic fusion concept after Yeh and Simunek (2002).

tions and prediction of flow phenomena using GPR and hydrological measurements collected during transient-flow experiments. They use the pilot point method (Certes and de Marsily, 1991; de Marsily, 1994) in a “maximum a priori” framework to generate saturated hydraulic conductivity distributions that are conditional to point measurements, that maintain specified patterns of spatial correlation, and that are consistent with geophysical and hydrological data. They found that inversion with different data types, including transient hydrological and GPR measurements, allowed good prediction of flow phenomena.

Data fusion methods extensively use local-scale information on state variables and parameters characterizing soil water processes. At present, there are two groups of methods available that allow us to quantify spatially variable properties and state variables in a space continuum. The first group of methods consists of hydrogeophysical and geophysical techniques (Vereecken et al., 2003, 2005; Rubin and Hubbard, 2005). At present, most of these studies have been undertaken in groundwater systems, with some studies limited to synthetic case studies. Recently, attention has been given to the application of these methods in the vadose zone (Yeh et al., 2002; Vereecken et al., 2004, 2006; Liu and Yeh, 2004). The most frequently used methods include GPR to derive spatial information on soil moisture content and hydraulic properties (Binley et al., 2001, 2002; Huisman et al., 2001, 2003; Annan, 2002; Lambot et al., 2004) and ERT techniques to derive soil moisture distribution and effective parameters (e.g., Daily et al., 1992; Zhou et al., 2001; Binley et al., 2002). Often GPR is used in combination with either ERT (Binley et al., 2002; Day-Lewis et al., 2005) or seismic methods (Chen et al., 2001) to study the dynamics of soil moisture. The second group of methods are remote sensing techniques (see above). At present, however, information and data obtained from this class of methods have not been used in the data fusion concept.

Data fusion provides a powerful concept to integrate different types of information characterizing soil water processes and available upscaling theories (e.g., stochastic perturbation theory) into an inverse stochastic framework. It provides a way to fully exploit spatially distributed information obtained from either hydrogeophysical or remote sensing methods. We expect that combining remote sensing techniques with hydrogeophysical methods using stochastic data fusion methods will significantly contribute to improving and validating methods of upscaling soil water processes and hydraulic parameters, especially in field-scale applications. Moreover, both remote sensing and hydrogeophysical techniques provide the possibility to monitor state variables (e.g., water content) and fluxes at the soil–plant–atmosphere boundary (e.g., root water uptake and evapotranspiration) in a minimally invasive fashion and under natural conditions. Applications of these methods may improve the quantification of vegetation controls on the subsurface environment and lead to an improved understanding of the spatial variability of soil water dynamics at the field scale. Only a few studies have

been conducted that explicitly account for the effect of vegetation on these processes and the upscaling process (Sharma and Luxmoore, 1979; Rubin and Or, 1993; Zhu and Mohanty, 2004; Katul et al., 1997; Kim et al., 1997). Limitations may arise from the fact that data fusion methods rely strongly on measurement techniques that provide a good spatial and temporal resolution. Application of this type of approach may therefore become expensive. Moreover, there needs to be a good correlation between the electricomagnetic properties and the soil water properties of the vadose zone to fully exploit the information contained in hydrogeophysical and remote sensing data. At present, most of the data fusion methods use the concepts of stochastic perturbation methods for the forward calculation of the state variables. Therefore, some of the limitations (e.g., mildly heterogeneous flow domains and linearization) of these methods also apply to data fusion.

GENERAL CONCLUSIONS AND RESEARCH NEEDS

We reviewed different approaches to upscaling soil water processes from the local to the fields scale and to deriving effective hydraulic properties. Two major groups were distinguished: forward upscaling methods requiring explicit characterization of local-scale hydraulic properties, and inverse upscaling methods that use temporal and spatial information of soil water state and system variables as well as available “soft” information that describes the system and is useful to quantitatively constrain the model parameter space. In both approaches, information about the spatial structure of hydraulic properties and the temporal and spatial structure and variability of state variables and boundary conditions is important. Lack of such information is the most limiting factor in validating and applying any of these approaches. Only a limited number of field studies are presently available in the literature to validate these approaches, while a large number of numerical and analytical studies have been published. The use of hydrogeophysical methods combined with ground-based remote sensing techniques may contribute to resolving this issue and lead to the further development and validation of upscaling techniques. Application of remote sensing techniques, however, will require that additional processes such as energy fluxes will be coupled with water flow.

At present, most of the available upscaling approaches for soil water processes ignore the effects of vegetation at the land–atmosphere interface. There exists a need to develop upscaling approaches that explicitly account for the effects of growing plants under natural field conditions. This will require a better characterization of the structure and function of the aboveground and root systems of vegetation. Application of the knowledge presently available within the vadose zone community with respect to upscaling hydraulic properties and state variables into other fields of research such as global change, climate change, and irrigation agriculture will require the study of soil–plant–atmosphere processes and not only bare soils.

Both groups of upscaling methods show that the estimated effective parameters do not depend only on the properties of the heterogeneous flow field but also on boundary conditions. Nonhomogeneous, nonlinear interactions between flow processes across space and time within the vadose zone create and destroy net upscaled fluxes not captured by simple averaging or first-order upscaling procedures that are mathematically tractable. It is unclear whether an analytical effective field- or watershed-scale unsaturated flow equation can indeed be found that is consistent with flow processes at the Darcy scale and measurements taken at various scales. The use of the Richards equation appears to be based more on pragmatism and lack of understanding of processes governing the flow of water at larger scales than on a sound physical basis.

Research on the upscaling of soil water processes and hydraulic properties in the field of vadose zone research has, in the past, focused strongly on the use of stochastic perturbation theories. Exploring the potential of volume averaging methods, homogenization methods, and renormalization methods in deriving effective properties of vadose zone properties has not yet been undertaken and fully accounted for. Application of these methods to the specific needs and questions of vadose zone hydrology and their comparison with the forward upscaling methods presented here might be a worthwhile task to undertake.

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